When Does a Small Model Know to Hand Off? Reading Competence from Full-Duplex Speech Models for Real-Time Escalation

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Abstract

Real-time speech systems often rely on compact models, but some user queries require a stronger expert. Selective escalation is well studied for text, yet speech adds a deployment question: does a duplex-trained model expose a reliable competence signal before it commits to a response, and does that signal survive distribution and modality shift? We compare decode-time uncertainty, verbalized self-assessment, and linear probes across layers and matched model controls, using leave-one-query-pool-out evaluation, matched text and speech inputs, and end-to-end escalation loops. On MiniCPM-o 4.5, an in-distribution final-layer probe scores AUC 0.822 but falls to 0.372 on held-out math with text input; the same last-token read at layer 22 reaches 0.931, while the cause of the late-layer failure remains unresolved. A text-calibrated layer-22 probe reads speech-input states at AUC 0.855, and a single-position end-of-turn study reaches 0.887 in $\sim$30 ms, before the first output token. The live system uses the same layer with a separately calibrated three-position probe. In the primary turn-based, audio-input/text-output evaluation, selective escalation raises deployed-channel answer accuracy from 42.2% to 63.3% at 57% escalation and beats matched-rate random escalation by 7.2–9.5 points; a separate TTS-on replay leaves gate decisions unchanged when speech synthesis is enabled. With its weights, features, and layer fixed, the gate also improves all five external speech-QA benchmarks over always-local operation, although it does not beat matched-rate random escalation on every pool. These results identify a mid-network representation, rather than the conventional final-layer read, as a practical signal for handoff before response commitment.

1 Introduction

Real-time spoken interaction leaves only a few hundred milliseconds between turns [Stivers et al., 2009]. Systems that meet this deadline often rely on a compact interaction model, but no compact model can answer every long-tail factual, technical, or multi-step question. A stronger cloud model can handle some of these turns, provided the system knows when to call it. In a text interface, this is a standard routing problem. In a speech interface, the decision must also arrive before the local talker commits to a response; otherwise the system must delay the expert call or retract an answer already under way. We keep the local speech model as the talker so that the audio context and voice remain in one session. This setting raises the central question of this paper: *does a duplex-trained speech model expose its own likely failure early enough to hand the turn to a stronger model?*

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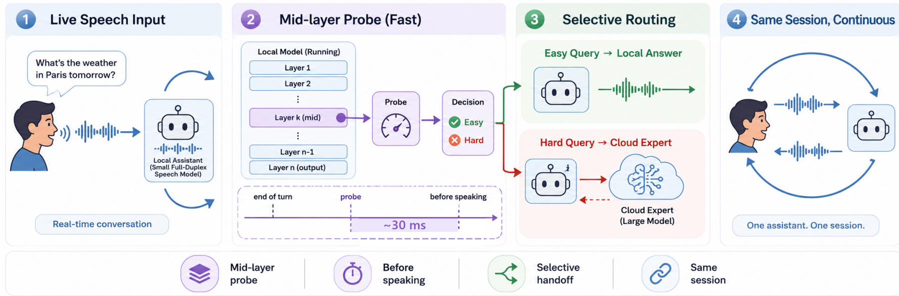


Figure 1: **Primary turn-based system.** **(1)** Speech reaches a duplex-trained local model. **(2)** At the end of the user turn, a mid-network (L22) linear read scores the turn in ∼30 ms, before the first output token. **(3)** Lower-risk turns stay local; higher-risk turns are sent to a cloud expert. **(4)** The local model remains the talker and relays the returned answer. Deployed-channel accuracy rises from 42.2% to 63.3% at 57% escalation in the audio-input/text-output sweep; a TTS-on replay validates the speech path separately.

Two lines of work provide only part of the answer. Text-language-model studies estimate correctness from verbal self-assessment, output uncertainty, or hidden states [Kadavath et al., 2022, Azaria and Mitchell, 2023, Burns et al., 2023, Kossen et al., 2024]; text routers then use query-side or pre-generation features to choose a model [Ong et al., 2024, Varshney et al., 2026, Lugoloobi et al., 2026]. Speech-model research instead focuses on when a duplex model should listen, speak, or invoke a trained thinking policy [Wang et al., 2024a, Fang et al., 2026, Huang et al., 2026]. These results do not establish whether a frozen duplex-trained checkpoint retains a competence read that survives both distribution shift and speech input, or whether such a read is available before response commitment.

Three linked conditions make this gap operational. First, a useful score must distinguish individual failures rather than memorize the type of query seen during calibration. Second, a score calibrated on inexpensive text data must remain meaningful when the same content arrives as audio. Third, it must be available early enough to launch the expert and must improve which turns are escalated at a fixed call budget. A signal that works only in distribution, only on text, or only after a full draft cannot support real-time handoff.

We test these conditions on MiniCPM-o 4.5 [Cui et al., 2026], a 9B omni model with full-duplex training. We compare decode-time uncertainty, verbalized self-assessment ($p(\mathrm{True})$), and linear hidden-state probes across layers and positions. Matched controls span raw backbones, streaming-omni and audio-only fine-tunes, a frozen-backbone audio adapter, and a second duplex generation. The controlled study uses judged failure labels on 600 public-benchmark queries in five frozen pools, with leave-one-pool-out (LOPO) transfer as the primary robustness test and matched text and speech inputs for modality transfer. We then measure when the score becomes available, insert it into a complete escalation loop, and freeze its layer and features for evaluation on public speech benchmarks. Model checkpoints remain frozen; only logistic probes are fit on stored activations. Throughout, *duplex-trained* describes the weights. The primary live evaluation uses a turn-based, end-of-turn decision with audio input and text output. A TTS-on replay measures speech onset, while separate experiments test concurrent and native full-duplex regimes (Appendices H and O).

The results form one evidence chain:

1. **Transfer changes where competence can be read.** The conventional final-layer probe scores AUC 0.822 in distribution but falls to 0.372 on held-out math with text input. On the same LOPO test, the last-token read at L22 reaches 0.931. Matched comparisons localize the late-layer failure to the duplex checkpoints we test, but direct controls do not identify its mechanism (Appendix D).

2. **The mid-layer read survives speech and arrives before response commitment.** A text-calibrated L22 probe reads speech-input states at AUC 0.855, with the same mid-layer band reproduced on real human recordings. In a single-position read-point study, the end-of-turn read reaches AUC 0.887 in $\sim$30 ms, before the first output token. A separate TTS-on replay leaves the gate decisions unchanged and confirms that the read precedes synthesized speech.
3. **The early read improves selective escalation.** A separately calibrated three-position probe at the same layer drives the primary audio-input/text-output live loop. Deployed-channel accuracy rises from 42.2% to 63.3% at 57% escalation and exceeds matched-rate random escalation by 7.2–9.5 points. With its layer and features frozen, the gate improves all five external speech-QA benchmarks over always-local operation, although selectivity over random is limited on some pools. The native duplex regime also works after regime-specific recalibration.

The central result is therefore narrower than a mechanism claim: the model does not explain why it will fail, but its likely failure becomes more reliably readable mid-network than at the conventional output interface, early enough to support a live handoff.

2 Related Work

Pre-answer competence estimation. Text-language-model correctness can be estimated from verbal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], output uncertainty, or hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024]. Intermediate layers can carry more useful information than the output distribution [Orgad et al., 2025], and trained hidden-state probes are strong factual-confidence estimators [Mahaut et al., 2024]. Semantic entropy improves uncertainty estimation by comparing sampled answers [Kuhn et al., 2023, Farquhar et al., 2024]; Semantic Entropy Probes amortize this signal into a pre-generation or post-hoc hidden-state read [Kossen et al., 2024]. Attention-map probes and multimodal uncertainty fusion extend the same broad idea [Chuang et al., 2024, Zhang et al., 2026a]. This work establishes that competence is readable in text models. It does not show where the read remains reliable after duplex speech training or whether a probe calibrated on text transfers to speech-input states.

Selective routing. HybridLLM [Ding et al., 2024], RouteLLM [Ong et al., 2024], and query-side routers [Xue et al., 2025, Hu et al., 2024, Li et al., 2026] choose a model for a complete text query from query features or preference data. FrugalGPT organizes cascades that can score an earlier model's response [Chen et al., 2023], while AutoMix verifies a drafted answer before escalating [Aggarwal et al., 2024]. Closest to our signal, recent methods read pre-generation activations or fit linear success probes on frozen states [Varshney et al., 2026, Lugoloobi and Russell, 2025, Lugoloobi et al., 2026]. A second family trains the decision into the generator: Toolformer and Self-RAG learn tool or reflection tokens [Schick et al., 2023, Asai et al., 2024]; Co-LLM and CITER learn token-level deferral [Shen et al., 2024, Zheng et al., 2025]; and pause, thought, or think/no-think tokens allocate more internal computation [Goyal et al., 2024, Zelikman et al., 2024, Yang et al., 2025, DeepSeek-AI et al., 2025]. Learning-to-defer and early-exit methods make related decisions within a model [Madras et al., 2018, Schuster et al., 2022]. Our contribution is not the linear pre-generation probe by itself. We test whether this kind of read survives duplex training and audio input, and use a deployment-time threshold without modifying the speech checkpoint.

Duplex control and live handoff. End-to-end spoken dialogue includes native duplex models [Nguyen et al., 2023, Défossez et al., 2024, Zeng et al., 2024, Fang et al., 2026], streaming omni models [Xu et al., 2025, Cui et al., 2026], frozen-backbone adapters [Wang et al., 2024b], and audio fine-tunes [Chu et al., 2024]. Their control interfaces decide when to listen or speak through state, idle, synchronization, or interruption tokens [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024, Ma et al., 2024, Fu et al., 2024, Yu et al., 2024], structured action channels [Zhang et al., 2026b], or small classifier heads [Liao et al., 2025, Chen et al., 2025]. MiniCPM-o 4.5 likewise predicts a listen/speak state before content generation [Cui et al., 2026]. These policies answer *when* the model should speak, not whether its content is likely to be correct.

Recent systems also connect a duplex interface to a stronger resource. MoshiRAG triggers asynchronous retrieval from the model's emerging output and returns knowledge to the local model [Chien et al., 2026]. DuplexOmni uses a trained think-control token to hand work to a separate reasoning layer [Huang et al., 2026]. These systems demonstrate that a live handoff is possible, but they do not evaluate a competence score read from a frozen duplex checkpoint before generation.

Positioning. Prior work therefore establishes three ingredients separately: competence can be read from text models, text queries can be routed, and duplex models can coordinate speaking or external computation. To our knowledge, the reviewed literature does not establish their conjunction: whether a frozen duplex-trained speech model exposes a pre-answer competence signal that survives unseen query pools and text-to-speech transfer, and whether that signal arrives early enough to improve live selective escalation. We evaluate these conditions directly. The claims are about readout robustness and system utility; the training mechanism behind the late-layer failure remains open.

3 Experimental Setup

Target model and matched-pair controls. The system target is **MiniCPM-o 4.5** (9B; omni + full-duplex fine-tune of Qwen3-8B), bf16, greedy decoding, seed 42. To attribute effects to the *kind* of fine-tune rather than the backbone, we assemble ten models spanning raw backbones, a second duplex generation (MiniCPM-o 2.6), a streaming-omni fine-tune (Qwen2.5-Omni-7B), an audio-only fine-tune (Qwen2-Audio-7B), and a frozen-backbone adapter model (Freeze-Omni, LLM weights byte-identical to its raw control); the full roster is Table 2 (appendix). The comparison statistic is always the within-pair difference.

Query pool, labels, and audio arm. 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test): *trap* = SimpleQA [Wei et al., 2024] (the target fails 100%), *hard-knowledge* = MMLU-Pro [Wang et al., 2024c], *easy-fact* = TriviaQA [Joshi et al., 2017], *easy-chat* = Dolly/Alpaca [Conover et al., 2023, Taori et al., 2023], *hard-math* = GSM8K tail + MATH-500 [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]; public benchmarks only, no LLM-generated queries. Adequacy labels come from an LLM judge [Zheng et al., 2023] with structured outputs, blind to the gate and validated by a stronger re-judge (agreement 96.2% text / 94.5% audio; every headline number moves $\leq$1 point under the stronger labels); the escalation expert is `gpt-5.5` at low reasoning effort. The audio arm renders the same frozen pool to speech (single-voice TTS, content unchanged — only modality varies), following Spoken-SQuAD / VoiceBench practice [Li et al., 2018, Chen et al., 2024]; audio-side findings are additionally validated on real human recordings (SD-QA, Faisal et al., 2021) and on the external benchmarks' official audio (§7). Judge prompts, pins, and spend: Appendix M–N.

Candidate competence signals. All model checkpoints remain frozen. We compare (i) decode scalars: max token entropy or margin over the first k decode steps; (ii) linear probes: logistic regression on hidden states, with layer and position selected on calibration data; and (iii) verbalized self-evaluation [Kadavath et al., 2022]: $p(\text{True})$-pre asks "would you answer this correctly?" before answering, $p(\text{True})$-post asks about a drafted answer; $P(\text{Yes})$ is read from the first token's distribution. The representation analyses use a single-position, 4096-d L22 probe fit on the 360-query calibration split. The live system uses the same layer plus two pooled positions, fit on a wider 2,310-query calibration mix that is disjoint from the external benchmarks. Concurrent and native serving change the read point, so those validations refit the same feature set in regime; Appendix B records the full calibration ledger.

Evaluation logic. In-distribution AUC is diagnostic, not the main robustness claim, because query-pool identity is predictive in the calibration mix. We therefore use LOPO AUC to test distribution transfer and text-trained/audio-scored AUC to test modality transfer. For the live system, the key comparison is against matched-rate random escalation: an improvement over always-local operation can come from the expert alone, while an improvement over random at the same call rate measures the value of the gate's ranking.

4 A Transferable Competence Read Survives Mid-Network

4.1 In-distribution accuracy hides a query-type shortcut

Table 3 compares the candidate signals on MiniCPM-o 4.5. In distribution the final-layer probe looks healthy at AUC 0.822 — but that number is mostly a *query-type* shortcut. The same hidden state all but names which pool a query came from, a pool-identity oracle alone recovers most of the probe's in-distribution margin. Appending true pool labels to the probe's features adds nothing because pool identity is already encoded. We therefore treat leave-one-pool-out (LOPO) transfer as the primary metric. With text input, the final-layer read falls to AUC 0.372 on held-out math under LOPO. When the trap pool is held out, its mean failure score is 0.232 even though the model fails every query in the pool. A query-surface router shows the same dependence on pool structure: it does not exceed the pool oracle in distribution and also degrades under LOPO, with little improvement from substantially more data (Appendices B, E). These results show why in-distribution routing accuracy alone cannot establish instance-level transfer.

Verbalized self-assessment transfers differently. Asked *before* answering, $p(\mathrm{True})$ catches exactly the trap pool the probe misses, scoring it 0.945. But timing is everything: asked *after* drafting, the model endorses its own confident-wrong answer and the signal inverts. Post-draft $p(\mathrm{True})$ is the strongest single readout in distribution, at 0.899, but it no longer provides a pre-answer routing signal.

4.2 Late-layer brittleness appears on text input in duplex checkpoints

The matched-pair pattern ties the failure to the duplex checkpoints rather than to the backbone alone. The raw Qwen backbone, subsampled to match MiniCPM-o 4.5's per-pool failure rates, holds LOPO math at 0.962–0.968 across five seeds where the duplex checkpoint scores 0.372. The Qwen2.5 family shows the same ordering: the raw backbone scores 0.809 (0.825 after failure-rate matching), the streaming-omni model scores 0.745, and the duplex MiniCPM-o 2.6 scores 0.538. The Qwen2-Audio comparison is not attribution evidence because the audio fine-tune also changes task capability and the failure distribution substantially (Appendix B). A layer $\times$ position sweep further localizes the difference (Figure 2): it is concentrated at the last-token position in late layers on text input. Mean-pooled reads remain usable at the same depth but trail the mid-layer peak by 0.13–0.15. At L22, the last-token probe reaches **0.931** on the same LOPO math test, close to the raw backbone.

The failure is input-specific rather than universal: with audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. We use L22 because one read point must remain robust for shifted text queries and support text-to-speech transfer, not because the final-layer signal is absent on the model's native audio input.

The sweep establishes *where* the transferable read survives, not *why* it weakens late. Direct tests do not support the simple hypothesis that turn-control state takes over the last-token representation: the layer at which performance drops does not move in speak mode, and the inverting probe direction aligns with neither the control vocabulary nor an explicit listen/speak direction. A decomposition further shows that a distributed transferable component weakens over the final layers of the two duplex checkpoints but not their raw backbones. These observations constrain the explanation but do not identify a training objective or causal mechanism; the full audit is in Appendix D.

The mid-network pattern also appears outside the MiniCPM family. On NemotronLabs-VoiceChat-11B, a hybrid-Mamba duplex model, the same recipe peaks at L30–34 of 56 and transfers to the external pools at AUC 0.70–0.79 with one quarter of the deployed probe's calibration data (Appendix C).

Figure 2: The core representational result: LOPO math AUC by layer and position. On the duplex model the standard (final-layer, last-token) read inverts on text input while mid-network layers hold $\geq$0.9; raw backbones show no cliff. The deployed gate reads L22.

4.3 Text-calibrated reads transfer to speech in end-to-end models

Probes trained on text hidden states and scored on speech-input hidden states locate where the representation becomes modality-shared: early layers are modality-specific (0.54–0.60), and from $\sim$50% depth text$\rightarrow$audio transfer plateaus at 0.82–0.87 (0.855 at the deployed L22). This replicates on MiniCPM-o 2.6 and cross-family on Qwen2.5-Omni. The decisive control is **Freeze-Omni**: identical frozen LLM weights with audio attached by an adapter — and cross-modal transfer remains at 0.52–0.60. This comparison supports a narrow operational rule: text calibration transfers in the end-to-end speech models tested here, but not in the adapter-attached control. Verbalized introspection shows the complementary failure: on audio input, pre-answer $p(\mathrm{True})$ collapses on both MiniCPM duplex generations while both non-duplex controls stay intact; controlled arms show the verbal self-check is sensitive to the text-token pathway — duplicating the question as text restores it — while a linear probe remains usable on the same forward pass (Appendix B). The mid-layer band also replicates on 200 real human recordings from SD-QA, so the result is not limited to the TTS rendering. The live system keeps L22 and adds two pooled positions to form the deployed probe (Appendix B).

5 The Read Arrives Before Response Commitment

Scoring latency. The gate is a linear read on L22 states the forward pass already computes — a 4096-d dot product per position, over one position (mechanistic probe) or three (the deployed probe adds two running means at no added scoring latency; Appendix B). On an H100, measured from prefill start with CUDA synchronization, the score is available at P50 in **20 ms for text and 45 ms for audio** (P95 25/104 ms). Time to first token on the same benchmark is 36/68 ms. The measurement excludes network time, but it establishes the ordering needed by the system: the cloud request can start before local decoding commits to an answer. Waiting for decoded tokens does not improve the read in our data. Max entropy over four tokens reaches AUC 0.696, and early-decode hidden states reach 0.776, compared with 0.828 at the prompt; waiting also postpones the expert request (Appendix F).

From score to escalation budget. On the frozen test split (n=240), always-local accuracy is 58.8% and always-expert accuracy is 91.7%. Thresholding the probe beats matched-rate random escalation at every operating point (Figure 7, appendix). At a 33% escalation budget, it reaches **77.9–78.7%**, recovering about 58% of the local-to-expert gap with one third of the expert calls. In distribution, the gate is statistically tied with a pool-identity oracle, as the shortcut audit predicts. Its advantage appears under transfer, where pool identity stops being sufficient (§4.1).

The gate is fit to *local failure*, not expert benefit: a benefit label would specialize the ranking to the current expert. The frozen scores remain useful under the stricter benefit label, but pools with little expert headroom remain a system-level limit (Appendix I). Comparisons with the model's built-in reasoning mode are in Appendix F.

6 Live End-of-Turn Escalation

Primary protocol. The protocol is turn-based: audio streams in 1 s chunks, but the talker speaks only after the user's turn ends and the model's native listen/speak head is not engaged. This protocol isolates competence handoff from turn-taking; the native-head validation is reported separately below. Scores from intermediate chunks are weaker than the offline read (best summary AUC 0.764) and can miss information in the last chunk. We therefore make one **end-of-turn read**: after the final audio chunk, the system prefills the assistant turn and reads L22 at its start. In the 360-query read-point study, the single-position probe reaches **AUC 0.887** in $\sim$30 ms, compared with 0.843 for the offline reference, and arrives before the first output token. This experiment establishes the read point. The accuracy sweeps below use the deployed three-position probe fit on the wider 2,310-query calibration mix (frozen-test AUC 0.879); tier thresholds are quantiles of its score.

If the turn is escalated, the local talker first produces a short stall phrase. Its own transcription of the input is sent to `gpt-5.5`, and the returned answer is injected as context for the local model to relay. The expert request runs asynchronously. Partial-query prefetch, expert reasoning effort, and injection controls are reported in Appendix G.

The live result. We ran 240 frozen test queries $\times$ four escalation tiers, each a complete turn-based loop with audio input. The primary accuracy sweep retains the model's output as text; a separate TTS-on run of the balanced tier verifies that enabling speech synthesis leaves all 84/240 gate decisions unchanged and measures speech onset. We score *deployed-channel answer accuracy*: correctness of the relayed content after the system path. Headline numbers use a pre-registered input filter that removes 21 formula-LaTeX queries and one broken recording, because the speech rendering destroys the symbolic input before the model receives it. Accuracy rises monotonically, $\mathbf{42.2}\% \rightarrow \mathbf{52.8}\% \rightarrow \mathbf{59.2}\% \rightarrow \mathbf{63.3}\%$ at 0/14/31/57% escalation, every tier's gain significant (paired bootstrap; Figure 12). More importantly, it exceeds the matched-rate random reference at every tier, by 7.2–9.5 points in the deployed view and by 5.4–8.4 points in the channel-controlled view (Figure 12). A measured always-escalate arm reaches 66.5%. The small remaining gain from 57% to 100% escalation is not evidence that selection is perfect: the same sessions reach 92.2% when the expert receives gold text, showing that the model's transcription uplink is the limiting channel at high escalation rates. Live accuracies have a replication floor of $\pm$2–3 points; no claim rests on a smaller difference. Full tables, latency profiles, and boundary cases are in Appendix G.

Serving-regime checks. A talker-on replay leaves the frozen ranking nearly unchanged (AUC 0.871/0.856 versus 0.866 headless). A teacher-forced concurrent interleave is less stable and requires an in-regime refit; its external results remain exploratory (Appendix H). Under the model's native full-duplex head, the read moves to the head's own listen$\rightarrow$speak commit point. Refitting the same features in that regime yields AUC 0.830 internally and 0.709 across the external QA pools. A native outcome re-mix beats matched-rate random on five of six pools. A full native live run on the internal 240 queries improves accuracy from 0.371 to 0.537 at 45% escalation, below the 0.596 re-mix prediction because relaying the returned answer is lossy. These checks support the ranking beyond the primary turn-based loop, but they also show that read points, probe weights, and thresholds must be calibrated for the serving regime (Appendix O).

7 Frozen Transfer to Public Speech Benchmarks

The query-type audit makes external transfer necessary. We freeze the probe weights, feature set, and layer selected on the internal calibration mix, then run the live loop of §6 on six public speech benchmarks with audio input and text output: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [Li et al., 2025] (spoken renderings of Joshi et al., 2017, Berant et al., 2013, and the set of Nachmani et al., 2024), SD-QA (real dialectal human speech) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All audio is the benchmarks' official pre-rendered audio; no evaluation query is used to fit the probe. The

Table 1: Transfer to five public speech-QA benchmarks. The top block is measured post-relay answer accuracy (%) from the audio-input/text-output live loop; the probe is fixed and thresholds use label-free per-pool quantiles. The bottom block applies the same recipe to NemotronLabs-VoiceChat-11B in an offline re-mix (Appendix C). Δ is the average gain over always-local. AlpacaEval, reported separately on its 1–5 scale, changes from 3.99 to 4.35 on MiniCPM but does not beat matched-rate random; the second family changes from 3.55 to 4.46 versus 4.23 for random. Full random references and gold-input ceilings are in Appendices I–J.

	TriviaQA	WebQ	Llama Q.	SD-QA	Reason. zh	avg	Δ
live post-relay view (audio input, text output)							
always-local	66.4	57.2	84.0	51.5	58.9	63.6	—
+ gate @15%	73.2	62.8	90.4	60.0	65.3	70.3	+6.7
+ gate @30%	80.0	68.0	90.4	72.0	71.3	76.3	+12.7
+ gate @50%	**86.0**	**73.2**	**94.8**	**79.5**	**77.2**	**82.1**	**+18.5**
always-escalate (measured)	88.0	80.8	93.6	88.5	83.7	86.9	+23.3
always-expert (gold-text ceiling)	96.8	86.4	92.8	93.0	87.1	91.2	+27.6
second family: NemotronLabs-VoiceChat-11B (offline re-mix, official judges)							
always-local	35.6	37.6	70.5	31.0	—	43.7	—
+ gate @15%	49.8	48.0	78.6	42.5	—	54.7	+11.0
+ gate @30%	61.5	58.8	83.8	54.5	—	64.6	+21.0
+ gate @50%	**74.1**	**69.6**	**87.6**	**67.5**	—	**74.7**	**+31.0**
matched random @50%	65.6	60.2	81.8	60.0	—	66.9	+23.2
always-expert (ceiling)	95.5	82.8	93.2	89.0	—	90.1	+46.4
probe AUC (frozen)	0.789	0.785	0.806	0.792	0.683	0.771	—
probe AUC (2nd family)	0.775	0.771	0.720	0.754	—	0.755	—
official offline	75.5	70.2	—	—	—	—	—

only per-pool adjustment is *label-free*: tier thresholds are quantiles of that pool's score distribution, requiring no correctness labels or judge calls. This controls the call budget but cannot improve ranking. A single absolute threshold also beats matched-rate random on four of five QA pools, but its fire rate varies from 2% to 48% (Appendix I). We use each benchmark's official judge when available and verify that the local-only controls are on the published scale.

The frozen scores remain discriminative outside the calibration mix: AUC ranges from 0.683 to 0.806 across the five QA pools, with a mean of 0.771 (Table 1). The gate also improves deployed-channel accuracy over always-local operation on every QA benchmark and at every budget; at 50% escalation, mean accuracy rises from 63.6% to 82.1%. These gains establish system utility but do not by themselves establish selectivity, because random escalation also adds expert answers.

Llama Questions provides the cleanest fixed-budget test. It contains a single query type, the kept-local subset scores 97.6% while the escalated subset scores 69.6% (z=6.46), and expert answers improve the latter to 88.8% (p<.0001). At 50% escalation, the measured system reaches 94.8%, compared with 93.6% for always escalating. Matched-rate precision reaches 95% of its base-rate cap (Appendix J). Cross-lingual transfer also improves when the English calibration mix is widened: Chinese Reasoning QA rises from AUC 0.621 to 0.683 despite adding no Chinese calibration data.

The transfer results also expose the boundary between predicting local failure and creating expert benefit. SD-QA is a positive real-speech case and exceeds matched-rate random at all three budgets. Web Questions is less stable, with little gain over random at the low and high budgets; in its top-scoring 15%, many local failures are also missed by the expert (Appendix I). AlpacaEval is a clearer negative: its open-ended failures do not present the retrieval-failure pattern the probe reads. On the second duplex family, which has more expert headroom on these pools, the same recipe beats matched-rate random on all five evaluated pools, including AlpacaEval ($p \leq .0004$). Thus external utility depends on both the quality of the competence ranking and whether the selected failures are repairable by the expert (Appendices C, I, and J).

8 Discussion

What the experiments establish. The three parts of the study answer different requirements of the same handoff decision. First, in-distribution probe accuracy is not enough: query type explains much of the apparent performance, and LOPO evaluation changes the preferred read from the final layer to the middle of the network. Second, this mid-layer read remains useful when text calibration is applied to speech-input states and is available before response commitment. Third, its ranking improves a live escalation loop over matched-rate random selection. External and native evaluations support the same conclusion, but also show that neither probe weights nor thresholds should be assumed to transfer across serving regimes without validation.

What the score represents. The probe is strongest on retrieval-like failures, where the model's state reflects a missing answer before generation. It is weak on false premises and complementary to signals that appear during decoding (Appendix G). This construct boundary explains why failure AUC and system benefit are not identical. A correct ranking cannot help when the expert fails on the same turns, and an open-ended quality benchmark may not contain the factual-retrieval event the probe detects. The unstable Web Questions curve and the negative AlpacaEval result are therefore part of the finding, not exceptions to hide; SD-QA, by contrast, confirms selective gains on real human speech. Selective escalation depends on both readable local failure and expert headroom.

Calibration is part of deployment. The model weights stay frozen, but the gate is still a fitted component. Text-only calibration is sufficient for the controlled cross-modal result on end-to-end speech models. The deployed three-position probe uses a wider benchmark-disjoint calibration mix, and both concurrent and native read points require in-regime refits. The method avoids fine-tuning the talker and lets the operating budget move through a threshold, but it does not eliminate the need for representative calibration data.

Limitations. (i) One phrasing per $p(\mathrm{True})$ variant; real-speech validation is scripted read speech in one dialect; formula-symbolic content is out of scope for the audio arm (any TTS render destroys it). (ii) The second-family replication covers mid-band readability and frozen-recipe transfer only — not matched-pair attribution, the live loop, or the concurrent regime. (iii) Labels come from one judge family (a stronger re-judge bounds deltas at $\leq$1 point), no human labels. (iv) One session per query per tier ($\pm$2–3 point floor); under concurrent prefill the frozen read degrades (AUC .87$\rightarrow$.76) and an in-regime refit recovers .82 — calibration must follow the regime, and scale helps only partly (a full 2,310-row in-regime refit lifts external mean AUC .64$\rightarrow$.69, still short of turn-based .77); that concurrent loop is exploratory (teacher-forced carrier, random-clearance internal but spotty externally; Appendix H). The deployed *native* full-duplex regime lands strictly between the two (in-regime AUC .83 internal / .71 external mean; Appendix O); its internal full live run is limited by a lossy relay channel. (v) Layer and features were chosen on internal calibration-split cross-validation and confirmed on held-out pools, not by nested selection inside each held-out pool (Appendix B). (vi) The mechanism behind the late-layer decay on text input is unresolved (Appendix D). (vii) The primary evaluation uses scripted, single-turn questions. Disfluent correction, chained tool calls, and multi-turn grounding remain open [Lin et al., 2026]; in the native test, a new question during the expert wait derails the pending relay in 7/10 trials, so production systems must cancel or replace an in-flight request when the user moves on.

Conclusion. In the duplex speech models studied here, the conventional final-layer read is not the most reliable place to estimate answer competence when text queries shift across pools. A mid-network read retains that query-pool transfer and remains usable when calibrated on text and scored on speech-input states. It also becomes available before response commitment and improves which turns a live system escalates. The result supports a practical handoff mechanism while leaving the cause of the text-input late-layer failure open.

References

Pranjal Aggarwal, Aman Madaan, Ankit Anand, Srividya Pranavi Potharaju, Swaroop Mishra, Pei Zhou, Aditya Gupta, Dheeraj Rajagopal, Karthik Kappaganthu, Yiming Yang, Shyam Upadhyay, Manaal Faruqui, and Mausam. AutoMix: Automatically mixing language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2310.12963.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.11511.

Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2304.13734.

Jonathan Berant, Andrew Chou, Roy Frostig, and Percy Liang. Semantic parsing on Freebase from question-answer pairs. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2013. WebQuestions.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2212.03827.

Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.

Qian Chen, Yafeng Chen, Yanni Chen, et al. MinMo: A multimodal large language model for seamless voice interaction. *arXiv preprint arXiv:2501.06282*, 2025.

Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li. VoiceBench: Benchmarking LLM-based voice assistants. *arXiv preprint arXiv:2410.17196*, 2024.

Chung-Ming Chien, Manu Orsini, Eugene Kharitonov, Neil Zeghidour, Karen Livescu, and Alexandre Défossez. MoshiRAG: Asynchronous knowledge retrieval for full-duplex speech language models. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2604.12928.

Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, et al. Qwen2-Audio technical report. *arXiv preprint arXiv:2407.10759*, 2024.

Yung-Sung Chuang, Linlu Qiu, Cheng-Yu Hsieh, Ranjay Krishna, Yoon Kim, and James Glass. Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. GSM8K.

Mike Conover, Matt Hayes, Ankit Mathur, Jianwei Xie, Jun Wan, Sam Shah, Ali Ghodsi, Patrick Wendell, Matei Zaharia, and Reynold Xin. Free dolly: Introducing the world's first truly open instruction-tuned LLM. Databricks blog, 2023. databricks-dolly-15k.

Junbo Cui, Bokai Xu, Chongyi Wang, Tianyu Yu, Weiyue Sun, Yingjing Xu, et al. MiniCPM-o 4.5: Towards real-time full-duplex omni-modal interaction. *arXiv preprint arXiv:2604.27393*, 2026.

DeepSeek-AI, Daya Guo, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.

Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-aware query routing. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2404.14618.

Fahim Faisal, Sharlina Keshava, Md Mahfuz ibn Alam, and Antonios Anastasopoulos. SD-QA: Spoken dialectal question answering for the real world. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021. arXiv:2109.12072.

Qingkai Fang, Shoutao Guo, and Yang Feng. BayLing-Duplex: Native full-duplex speech dialogue with a single autoregressive LLM. *arXiv preprint arXiv:2606.14528*, 2026.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.

Chaoyou Fu, Haojia Lin, Zuwei Long, Yunhang Shen, Yuhang Dai, Meng Zhao, et al. VITA: Towards open-source interactive omni multimodal LLM. *arXiv preprint arXiv:2408.05211*, 2024.

Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training language models with pause tokens. In *ICLR*, 2024.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks*, 2021. arXiv:2103.03874.

Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt Keutzer, and Shriyash Kaustubh Upadhyay. RouterBench: A benchmark for multi-LLM routing systems. *arXiv preprint arXiv:2403.12031*, 2024.

Muye Huang, Lingling Zhang, Xingyu Yu, Lei Shi, Zhanyu Ma, Jun Xu, Jiuchong Gao, Jinghua Hao, Renqing He, and Jun Liu. DuplexOmni: Real-time listening, seeing, thinking, and speaking for full-duplex interaction. *arXiv preprint arXiv:2606.09186*, 2026.

Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint arXiv:2406.15927*, 2024.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2302.09664.

Chia-Hsuan Li, Szu-Lin Wu, Chi-Liang Liu, and Hung-yi Lee. Spoken SQuAD: A study of mitigating the impact of speech recognition errors on listening comprehension. In *Interspeech*, 2018. arXiv:1804.00320.

Hao Li, Yiqun Zhang, Zhaoyan Guo, Chenxu Wang, Shengji Tang, Qiaosheng Zhang, Yang Chen, Biqing Qi, Peng Ye, Lei Bai, Zhen Wang, and Shuyue Hu. LLMRouterBench: A massive benchmark and unified framework for LLM routing. *arXiv preprint arXiv:2601.07206*, 2026.

Tianpeng Li, Jun Liu, Tao Zhang, Yuanbo Fang, Da Pan, Mingrui Wang, et al. Baichuan-Audio: A unified framework for end-to-end speech interaction. *arXiv preprint arXiv:2502.17239*, 2025. Introduces OpenAudioBench.

Borui Liao, Yulong Xu, Jiao Ou, Kaiyuan Yang, Weihua Jian, Pengfei Wan, and Di Zhang. FlexDuo: A pluggable system for enabling full-duplex capabilities in speech dialogue systems. *arXiv preprint arXiv:2502.13472*, 2025.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.20050; source of the MATH-500 subset.

Guan-Ting Lin, Jiachen Lian, Tingle Li, Qirui Wang, Gopala Anumanchipalli, Alexander H. Liu, and Hung-yi Lee. Full-Duplex-Bench: A benchmark to evaluate full-duplex spoken dialogue models on turn-taking capabilities. *arXiv preprint arXiv:2503.04721*, 2025.

Guan-Ting Lin, Chen Chen, Zhehuai Chen, and Hung-yi Lee. Full-Duplex-Bench-v3: Benchmarking tool use for full-duplex voice agents under real-world disfluency. *arXiv preprint arXiv:2604.04847*, 2026.

Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022. arXiv:2205.14334.

William Lugoloobi and Chris Russell. LLMs encode how difficult problems are. *arXiv preprint arXiv:2510.18147*, 2025.

William Lugoloobi, Thomas Foster, William Bankes, and Chris Russell. LLMs encode their failures: Predicting success from pre-generation activations. *arXiv preprint arXiv:2602.09924*, 2026.

Ziyang Ma, Yakun Song, Chenpeng Du, Jian Cong, Zhuo Chen, Yuping Wang, Yuxuan Wang, and Xie Chen. Language model can listen while speaking. *arXiv preprint arXiv:2408.02622*, 2024.

David Madras, Toniann Pitassi, and Richard Zemel. Predict responsibly: Improving fairness and accuracy by learning to defer. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Matéo Mahaut, Laura Aina, Paula Czarnowska, Momchil Hardalov, Thomas Müller, and Lluís Màrquez. Factual confidence of LLMs: on reliability and robustness of current estimators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2310.06824.

Eliya Nachmani, Alon Levkovitch, Roy Hirsch, Julian Salazar, Chulayuth Asawaroengchai, Soroosh Mariooryad, Ehud Rivlin, RJ Skerry-Ryan, and Michelle Tadmor Ramanovich. Spoken question answering and speech continuation using spectrogram-powered LLM. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.15255; Spectron, source of the Llama Questions dataset.

Tu Anh Nguyen, Eugene Kharitonov, Jade Copet, Yossi Adi, Wei-Ning Hsu, Ali Elkahky, Paden Tomasello, Robin Algayres, Benoît Sagot, Abdelrahman Mohamed, and Emmanuel Dupoux. Generative spoken dialogue language modeling. *Transactions of the Association for Computational Linguistics*, 2023. arXiv:2203.16502.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*, 2024.

Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.02707.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2302.04761.

Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2207.07061.

Shannon Zejiang Shen, Hunter Lang, Bailin Wang, Yoon Kim, and David Sontag. Learning to decode collaboratively with multiple language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. arXiv:2403.03870; Co-LLM.

Tanya Stivers, N. J. Enfield, Penelope Brown, Christina Englert, Makoto Hayashi, Trine Heinemann, Gertie Hoymann, Federico Rossano, Jan Peter de Ruiter, Kyung-Eun Yoon, and Stephen C. Levinson. Universals and cultural variation in turn-taking in conversation. *Proceedings of the National Academy of Sciences*, 106(26):10587–10592, 2009.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following LLaMA model. GitHub repository, 2023. zh split: shibing624/alpaca-zh.

Tanay Varshney, Annie Surla, Michelle Xu, Gomathy Venkata Krishnan, Maximilian Jeblick, David Austin, Neal Vaidya, and Davide Onofrio. LLM router: Rethinking routing with prefill activations. *arXiv preprint arXiv:2603.20895*, 2026.

Bandhav Veluri, Benjamin N. Peloquin, Bokai Yu, Hongyu Gong, and Shyamnath Gollakota. Beyond turn-based interfaces: Synchronous LLMs as full-duplex dialogue agents. In *Proceedings of EMNLP (Main)*, 2024. arXiv:2409.15594.

Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong. Freshllms: Refreshing large language models with search engine augmentation. *arXiv preprint arXiv:2310.03214*, 2023.

Peng Wang, Songshuo Lu, Yaohua Tang, Sijie Yan, Wei Xia, and Yuanjun Xiong. A full-duplex speech dialogue scheme based on large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024a. arXiv:2405.19487.

Xiong Wang, Yangze Li, Chaoyou Fu, Lei Xie, Ke Li, Xing Sun, and Long Ma. Freeze-Omni: A smart and low latency speech-to-speech dialogue model with frozen LLM. *arXiv preprint arXiv:2411.00774*, 2024b.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, et al. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks*, 2024c. arXiv:2406.01574.

Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. Measuring short-form factuality in large language models. *arXiv preprint arXiv:2411.04368*, 2024. SimpleQA.

Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, et al. Qwen2.5-Omni technical report. *arXiv preprint arXiv:2503.20215*, 2025.

Leyang Xue, Yao Fu, Luo Mai, and Mahesh K. Marina. HybridServe: Efficient serving of large AI models with confidence-based cascade routing. *arXiv preprint arXiv:2505.12566*, 2025.

An Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Wenyi Yu, Siyin Wang, Xiaoyu Yang, Xianzhao Chen, Xiaohai Tian, Jun Zhang, Guangzhi Sun, Lu Lu, Yuxuan Wang, and Chao Zhang. SALMONN-omni: A codec-free LLM for full-duplex speech understanding and generation. *arXiv preprint arXiv:2411.18138*, 2024.

Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah D. Goodman. Quiet-STaR: Language models can teach themselves to think before speaking. *arXiv preprint arXiv:2403.09629*, 2024.

Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong, and Jie Tang. GLM-4-Voice: Towards intelligent and human-like end-to-end spoken chatbot. *arXiv preprint arXiv:2412.02612*, 2024.

Dengjia Zhang, Alexander Martin, William Jurayj, Kenton Murray, Benjamin Van Durme, and Reno Kriz. Unified multimodal uncertain inference. *arXiv preprint arXiv:2604.08701*, 2026a.

Haoyang Zhang, Jun Chen, Donghang Wu, Yuxin Li, Yuxin Zhang, Xiangyu Tony Zhang, Che Liu, Qingjian Lin, Yizhou Peng, Hexin Liu, Eng Siong Chng, Chao Yan, Boyong Wu, Yechang Huang, Xuerui Yang, and Fei Tian. DuplexSLA: A full-duplex spoken language model with synchronized speech, language, and action. *arXiv preprint arXiv:2605.20755*, 2026b.

Xinrong Zhang, Yingfa Chen, Shengding Hu, Xu Han, Zihang Xu, Yuanwei Xu, Weilin Zhao, Maosong Sun, and Zhiyuan Liu. Beyond the turn-based game: Enabling real-time conversations with duplex models. In *Proceedings of EMNLP*, 2024. arXiv:2406.15718.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks*, 2023. arXiv:2306.05685.

Wenhao Zheng, Yixiao Chen, Weitong Zhang, Souvik Kundu, Yun Li, Zhengzhong Liu, Eric P. Xing, Hongyi Wang, and Huaxiu Yao. CITER: Collaborative inference for efficient large language model decoding with token-level routing. In *Conference on Language Modeling (COLM)*, 2025. arXiv:2502.01976.

A Model roster

Table 2: Model roster. "Pair" groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

B Extended signal and readout analyses

Table 3: Signals on MiniCPM-o 4.5, judged failure labels. In-dist = calibration-split AUC (probes 5-fold out-of-fold); LOPO math = leave-one-pool-out AUC on held-out math; audio = the same readout under speech input (final/mid-layer probe: audio-input LOPO math / text-only calibration read on audio states). Full per-signal detail: Appendix B.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	≈chance on raw backbones	—
$p(\text{True})$-pre	0.807	trap 0.945 (probe: 0.232)	**collapses**
$p(\text{True})$-post (full draft)	**0.899**	—	degrades
final-layer probe	0.822	**0.372** (inverts)	0.936 (intact)
mid-layer (L22) probe	0.866	**0.931**	**0.855** (text-calibrated)

The in-distribution shortcut, quantified. The numbers behind §4.1's claim that the final-layer probe's in-distribution AUC of 0.822 is mostly pool identity: a linear read of the same hidden state classifies a query's source pool at 95.8% accuracy; a pool-identity oracle alone reaches AUC 0.715; and appending true pool labels to the probe's features moves it 0.822→0.821. In-distribution area over random is statistically on par between the gate and the pool oracle (Appendix E); the two separate only under LOPO, where the probe inverts to 0.372 and the oracle is undefined.

ROC curves. Figure 3 shows the calibration-split ROC for the signals of Table 3.

$p(\text{True})$ **backbone dependence.** Across backbones, post-draft $p(\text{True})$ beats the trained final-layer probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe 0.798) shows the pattern depends on the backbone's self-evaluation quality.

Matched pairs. Table 4: with label coverage matched by subsampling the raw control to the duplex model's per-pool fail rates, the raw backbone keeps LOPO math at 0.825/0.809 while duplex models sit at chance — label coverage is ruled out; the representation changed. Degradation is ordered raw $>$ omni-streaming $>$ duplex. Table 5 gives the by-depth summary behind Figure 2: the within-pair deficit at the best mid-layer is only -0.03 vs. -0.59 at the final layer; mean-pooled probes survive to the final layer on duplex models (LOPO math ≈0.80 at L35 on o4.5, 0.674 at L27 on o2.6 — 0.13–0.15 below each model's mid-layer last-token peak, and uniformly weaker in-mix, -0.058 at L22); duplex damage is severe but local, omni-streaming damage mild but diffuse.

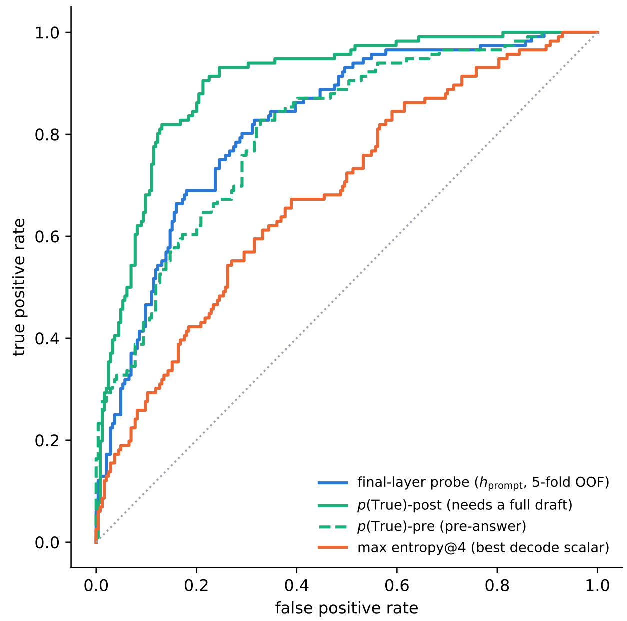


Figure 3: ROC on the calibration split. $p(\mathrm{True})$-post leads but needs a full drafted answer; the probes and $p(\mathrm{True})$-pre are pre-answer.

Table 4: LOPO transfer (AUC, held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6	[o4.5 vs Qwen3]
hard-math	**0.809**	0.745	**0.538** ≈ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	**0.526** ≈ chance	0.61

Modality specificity of the cliff. Under audio input the final layer does *not* invert (L35 audio LOPO math 0.936 vs. 0.366 text); a speak-mode template control on text input leaves the cliff unchanged (0.362) — the operative variable is the modality of the input context, not the output mode.

Audio collapse of verbalized introspection. Table 6. Three controlled arms pin the mechanism on MiniCPM-o 4.5: *not perception* (ASR audit WER 0.074; $p(\mathrm{True})$ on the model's own transcript snaps back to 0.074; the well-heard subset still collapses); *not a context prior* (irrelevant filler audio does not inflate p_{yes}); *binding* (duplicating the question as text alongside the audio fully restores introspection, 0.034). The verbal instance-check runs over text-token pathways that audio embeddings do not feed, while a linear probe reads the same information off the same forward pass at 0.93+. The failure implies its own cheap fix, confirmed twice: re-present the query as text before asking ("repeat-then-judge").

Real-speech validation. The matched-pair design rerun on 200 real human recordings (VoiceBench SD-QA, USA split) replicates all three audio-side findings: modality tax (0.400→0.450), audio overconfidence (paired p_{yes} shift $+0.089$; failure subset 0.415→0.581), and the layer structure (text-calibrated read on real-speech audio holds 0.76–0.80 at L22–L26, peak 0.800 at L25 — transfer across both content and modality).

Synthesis. End-to-end multimodal training *builds* the shared mid-layer core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither (probe readout text-fragile, verbal readout audio-fragile; in both cases knowledge survives and a readout breaks, and in both cases the breakage is duplex-fine-tune-specific).

Table 5: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	**0.931 (L22/36)**	**0.366**	cliff, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse depression
MiniCPM-o 2.6 (duplex)	**0.822 (L21/28)**	**0.540**	cliff

Table 6: Pre-answer introspection under audio input. Trap p_{yes} = mean stated probability of answering correctly on a 100%-fail pool (lower is better).

model	FT type	trap p_{yes}: text $\rightarrow$ audio	audio $p(\text{True})$-pre AUC
MiniCPM-o 4.5	duplex	0.055 $\rightarrow$ **0.556** (collapse)	0.786 (trap dead)
MiniCPM-o 2.6	duplex	0.196 $\rightarrow$ 0.345	**0.491** $\approx$ chance
Qwen2.5-Omni	omni-streaming	0.279 $\rightarrow$ 0.213 (intact)	0.727
Freeze-Omni	frozen + adapter	0.165 $\rightarrow$ 0.112 (intact)	0.612 (capability caveat)

The deployed probe, and which calibration each result uses. The mechanistic probe is a single-position L22 read (4096-d), fit on the 360-query calibration split; every representational result (§4, Appendices B, D) is this probe. The deployed probe adds two zero-latency pooled positions (an 8-token tail mean and the running user-audio mean; 12,288-d), refit on a calibration mix widened to 2,310 public-benchmark queries disjoint from every evaluation benchmark; a pre-registered guard held (frozen internal test AUC 0.860→0.879) and feature selection never saw the externals. It drives the turn-based live loop (§6), the frozen transfer results (§7), and the talker-on replay — frozen throughout, calibrated on text and speech-rendered internal data only. The serving-regime validations are the exceptions. In the concurrent regime, the same 12,288-d feature set is refit in-regime on 360 rows for the loop of Table 13 and on the full 2,310-row mix to measure the effect of calibration scale (Appendix H). In the native regime, the same feature set is refit on 2,310 rows collected at the native listen→speak read point (Appendix O). *Selection history.* L22 and the three-position feature set were chosen on internal calibration-split cross-validation before any external benchmark was scored (the 8-position $\times$ layer sweep and the 19-configuration feature sweep are both calibration-only), and the internal frozen test split was the pre-registered guard. The external pools were then read once with the frozen artifact. Two honest exposures remain: the layer band was picked on internal LOPO rather than by nested selection inside each held-out pool, and the second duplex family's mid-band peak (L30–34 of 56) was located with its own sweep, so both layer choices are best read as *confirmed on held-out pools*, not as untouched-holdout estimates. In the mid-layer band the in-mix tradeoff optimum is broad (L20–L30) and the mid-layer probe's in-distribution area advantage is modest — its real advantage is LOPO robustness, which an in-mix test cannot show.

C Second duplex family: NemotronLabs-VoiceChat-11B

The pre-registered prediction test: with the identical recipe (same audio, same judge, 600-query calibration), the layer sweep on this hybrid-Mamba duplex model (56 blocks: 27 Mamba-2 / 4 attention) peaks mid-band at L30–34 (eot-last AUC 0.714 vs. 0.693/0.682 at the endpoints); stacking the same three positions lifts OOF 0.714→0.761→0.790; the frozen-methodology probe transfers to the external pools at AUC 0.781/0.793/0.754/0.701 — the deployed-probe band reached with a quarter of the calibration data. Boundary notes: the backbone is much weaker at knowledge retrieval (striviaqa local 0.32 vs. MiniCPM 0.62, same judge) and English-only (the cross-lingual pool has no analog). The live-loop port and the trained-checkpoint ablation remain future work; the port is gated by the released inference path, not by the architecture. The only supported loading path (the NeMo Speech `nemotron-labs-voicechat` branch) runs the duplex loop *cacheless* — every 80 ms frame re-runs the full prefix, an $O(T^2)$ schedule that cannot hold the frame clock in real time — although the hybrid backbone's Mamba-2 state is exactly the $O(1)$ per-frame recurrence a streaming port would need; wiring stateful inference into the released duplex frame protocol is the engineering gap. (Offline, the same property is a convenience: the final frame's forward contains every position, so one hook captures the whole eot window.) Accuracy-side conclusions do not wait on the port: the re-mix arithmetic below tracks measured live arms at 0.011 mean deviation on MiniCPM, inside the $\pm$0.02–0.03 replication floor.

Escalation re-mix on every transfer pool. Completing the grid of Table 1 on this family: an offline re-mix (the same top-r reconstruction arithmetic that, on MiniCPM, tracks the measured live arms at 0.011 mean deviation) sends the top-r of each pool by NVDA probe score to the measured GPT-5.5 outcome and keeps NVDA's own answer for the rest, scored with each pool's official protocol (the OAB judge for the three OpenAudioBench pools — NVDA local floors 0.352/0.376/0.705 — our judge for SD-QA, VoiceBench 1–5 for AlpacaEval; the expert answers the heard transcript and is judged directly, so the relay-back channel is excluded — including the measured relay outcomes where they exist shifts tier points by $\leq$0.032). **Selective escalation beats matched-rate random on all five pools** (permutation $p \leq 0.0004$ at the 30% and 50% budgets, $p < 0.03$ at 15%; Table 7, Figure 4) — including Web Questions, SD-QA and AlpacaEval, the three pools that are flat or negative on MiniCPM (§7). Selectivity is nearly unchanged (AUC 0.72–0.78 under official labels); what changed is headroom — floors of 0.31–0.38 against MiniCPM's 0.51–0.66 — and on AlpacaEval the failure species: the terse English-only model fails open-ended queries *hard* (the probe-selected half scores 3.01 vs. 4.08 kept), where MiniCPM's open-ended failures are soft quality gradations a wrongness detector cannot see. The negatives are properties of the pool $\times$ backbone pairing, not of the signal.

Table 7: The frozen-recipe NVDA probe re-mixed at fixed escalation rates, official judge per pool, each against its own matched-rate random reference (same measured outcomes, escalation set drawn at random; identical at 0% and 100% by construction). AlpacaEval rows are VoiceBench 1–5 scores; others accuracy. Every pool clears random at all three budgets: $p \leq 0.0004$ at 30% and 50%, $p < 0.03$ at 15% (20k permutations, `scripts/18_nvda_random_check.py`).

pool	judge	n	0%	15%	30%	50%	100%
Speech TriviaQA	OAB	247	0.356	0.498	0.615	0.741	0.955
matched random				0.446	0.536	0.656	
Speech Web Questions	OAB	250	0.376	0.480	0.588	0.696	0.828
matched random				0.444	0.512	0.602	
Llama Questions	OAB	234	0.705	0.786	0.838	0.876	0.932
matched random				0.739	0.773	0.818	
SD-QA	ours	200	0.310	0.425	0.545	0.675	0.890
matched random				0.397	0.484	0.600	
AlpacaEval (1–5)	VB	199	3.55	3.83	4.13	4.46	4.92
matched random				3.75	3.96	4.23	

D What breaks at the output: mechanism audit

§4.2 reports *that* the final-layer last-token read inverts. This section reports what we could and could not establish about *why*. Everything below uses the Phase-5d all-layer prefill dumps (every decoder layer, last-token and mean-pooled, same 600 queries) and the identical probe protocol as the layer sweep: calibration split, L_2 logistic regression, LOPO trained on the four non-math pools and scored on held-out math. The reproduction check recovers the published L22 and final-layer numbers to $\leq$0.012 (`scripts/14-17_*.py`).

Four mechanisms ruled out. Table 8 lists them. (i) *Speak mode.* Prefilling the same text queries under the TTS/speak-mode template leaves the cliff exactly where it was (final-layer LOPO math 0.362 vs. 0.366 plain), so the readout is not being displaced by a prepare-to-speak state. (ii) *Modality*

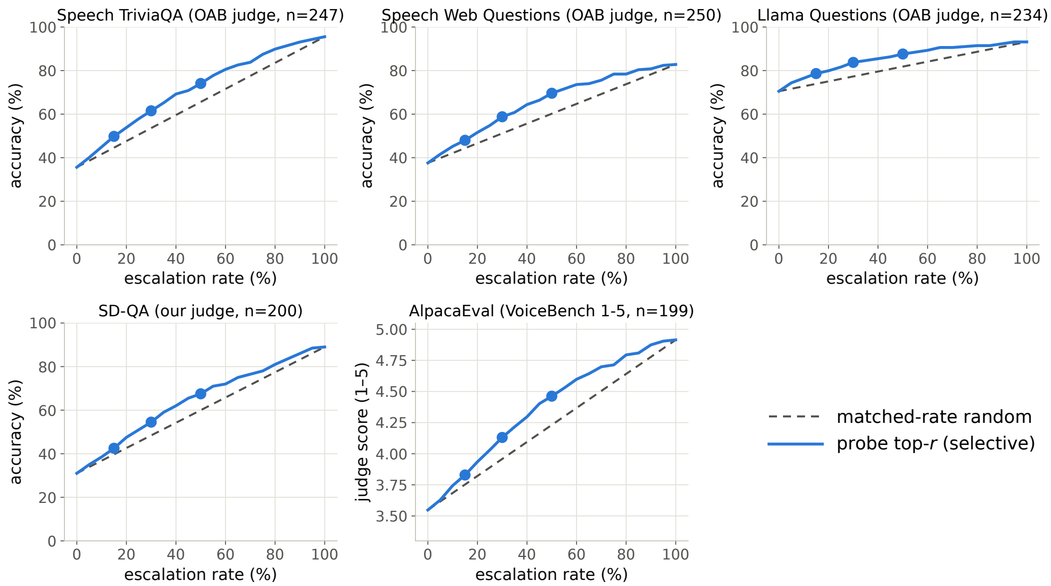
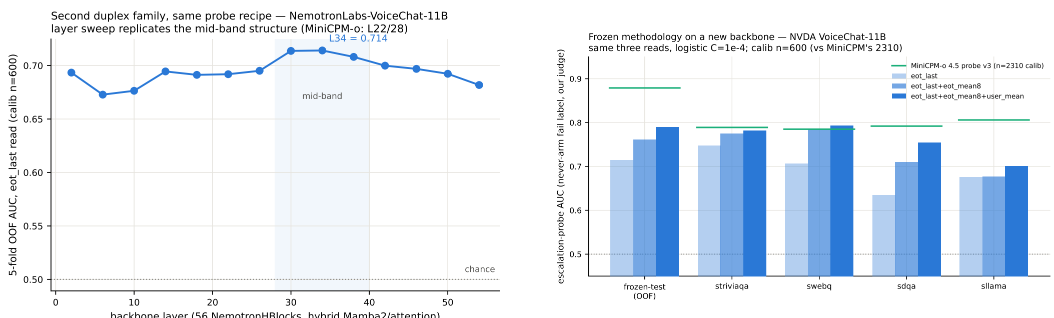


Figure 4: Re-mix curves for the frozen-recipe NVDA probe: selective (top-r by probe score) vs. matched-rate random on the five transfer pools, official judge per pool. Every pool separates, including the three that are flat on MiniCPM (all five pools, $p \leq 0.0004$ at 30/50% and $p < 0.03$ at 15%, permutation).



(a) layer sweep

(b) external transfer

Figure 5: Second-family test: (a) the usable read again peaks mid-network; (b) frozen-recipe transfer lands in the deployed-probe AUC band.

axis. The mean audio-minus-text direction at the final layer is orthogonal to both the inverting final-layer probe and the transferring L22 probe (cosine 0.013 and 0.027; chance is $1/\sqrt{4096} = 0.016$), so the cliff is not the audio/text split that §4.3 measures. (iii) *More shortcut available late.* Source-pool identity is linearly decodable at 0.88–0.96 at *every* depth, in the duplex model and its raw backbone alike — the query-type shortcut of §4.1 is not something the late layers add; it is what remains once the competence signal stops being readable. (iv) *Wholesale representational drift.* Raw linear CKA between the duplex model and its backbone appears to collapse late (0.80→0.47), but transformer residual streams carry a few massive-activation coordinates that dominate the Gram matrix; with per-feature standardization the same comparison is nearly flat (0.82→0.78) and does not localize the cliff. Projecting out the massive-activation axes, or the pool-mean directions, rescues nothing (0.35–0.40 at every $k \leq 20$, against a random-direction control of 0.37).

What does separate the models. Figure 6 measures the probe rather than operating on the model. For the probe trained at each depth on the four non-math pools, we split its held-out score into the part carried by that layer's five dominant principal directions (estimated from training rows only) and

Table 8: Mechanism candidates for the final-layer inversion. Final-layer LOPO hard-math AUC on MiniCPM-o 4.5 unless noted; 0.366 is the unmodified baseline and 0.931 the L22 read.

candidate	test	verdict
prepare-to-speak state	speak-mode template prefill	0.362, no change
modality (audio vs. text) axis	cosine with the probe direction	0.013 $\approx$ chance
late layers encode more pool identity	pool decoding by depth	0.88–0.96 at all depths
representation rewritten late	standardized CKA vs. backbone	0.82$\rightarrow$0.78, flat
massive-activation coordinates	project out top-k such axes	0.35–0.40, no rescue
query-type directions	project out pool-mean directions	0.34–0.45, no rescue
delivery to the output position	**residual component of the read**	**0.93$\rightarrow$0.27 (Fig. 6)**

the residual, $s = w^\top Px + w^\top(I-P)x$. The residual component is the distributed read, and it is the one that transfers: it holds 0.85–0.93 through mid-network on MiniCPM-o 4.5 and then decays to **0.27** at the output, with the same decay on MiniCPM-o 2.6 (0.81$\rightarrow$0.52) and *no* decay on either raw backbone (Qwen3-8B 0.86, Qwen2.5-7B 0.71 at their final layers) or on the omni-streaming control (Qwen2.5-Omni holds 0.75 to its output with a flat variance share) — the decay tracks duplex fine-tuning specifically, not streaming or audio fine-tuning in general. On MiniCPM-o 4.5 the score also concentrates: the dominant subspace carries 39% of the held-out score variance at the final layer against $\leq$0.18 at every earlier depth and 0.05 for its backbone; on MiniCPM-o 2.6 this concentration is present but weak (0.16 vs. 0.14 for its backbone), so we report it as a property of the stronger duplex model, not a general law.

Direct turn-control tests. Two experiments target the turn-control story head-on rather than by exclusion (`modal_interp.py`, `scripts/19_*.py`). *Control-token logit lens:* applying each model's own final norm and unembedding to the stored per-layer states, the probability mass on the trained-in control vocabulary (added/special tokens) *rises* over MiniCPM-o 4.5's last layers (log-mass -23.8 at L32 to -11.8 at the output) while the raw backbone's falls by fifteen log-units over the same span — the duplex output layer, uniquely, stays ready to emit control tokens. But the fine-grained prediction fails: the inverting probe direction is no more aligned with the control tokens' unembedding rows than with random vocabulary rows (max $|\cos|$ 0.069 vs. 0.068), and per-query control mass correlates with the probe score at only r=0.18. *Listen/speak intervention:* re-prefilling 60 calibration queries with a still-listening vs. answer-now suffix moves the final-layer state substantially in *both* models (relative displacement 0.27 duplex, 0.59 raw — cue text is just prompt semantics), and the cue direction is orthogonal to the inverting probe ($|\cos|$=0.019, chance 0.016), shifting its score by only $+0.23$ sd (and the deployed L22 read by $+0.08$ sd). Together: the duplex output layer is measurably control-biased in its output distribution, but the inversion is not carried by control-token directions or by an explicit listen/speak axis — the simple takeover story fails its own direct tests.

What we do not claim. Destructive tests do not settle the mechanism. Removing the top five principal directions from the final layer does raise LOPO math from 0.378 to 0.758 — but the identical operation costs the raw backbone 0.90$\rightarrow$0.14 and costs the intact L22 read 0.931$\rightarrow$0.760, so it is a damaging intervention that happens to help a broken readout, not evidence of a duplex-added subspace. We therefore state the finding at the level the evidence supports: duplex fine-tuning does not erase the competence signal, it stops delivering it in distributed form to the final last-token position, and the layer where delivery fails is also the one whose output distribution stays control-biased. Attributing the loss to a specific duplex objective needs training-time access we do not have; the practical consequence — read mid-network — does not depend on the attribution.

E Query-surface router audit

A RouteLLM-style TF-IDF$\rightarrow$LR router under identical labels reaches OOF AUC 0.669 (below the pool oracle's 0.715); a 24-configuration sweep tops out at 0.689, so this is the query surface's ceiling,

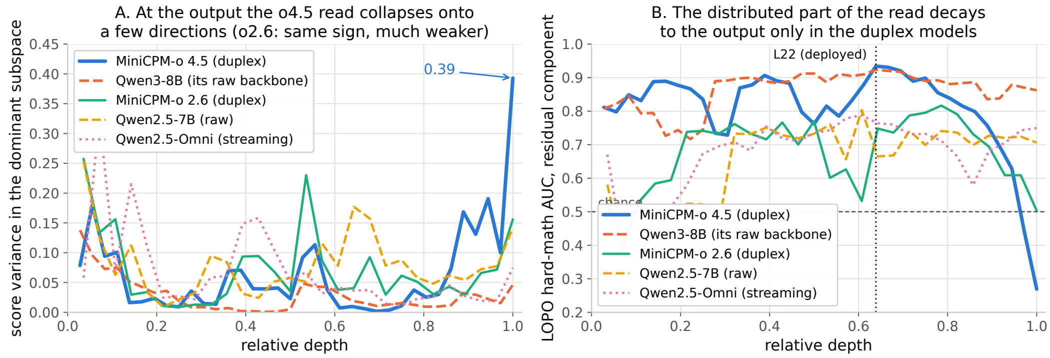


Figure 6: Non-destructive decomposition of the probe's held-out score at every depth. **Left:** share of score variance carried by the layer's five dominant directions. **Right:** transfer AUC of the residual (distributed) component — intact to the output on both raw backbones and on the omni-streaming control, decaying to inversion over the final layers of both duplex models.

not tuning. Under LOPO it collapses to 0.38–0.57 and scores the 100%-fail trap pool at 0.230 — it misses the pool a router most needs. The training receipt: at the argmax threshold its out-of-fold accuracy exactly matches the majority class; the internal probes under the same accounting clear majority by 9–29 points (Figure 8). Nor is it data-starved: the identical recipe on RouterBench [Hu et al., 2024] (36,497 prompts) reaches only in-domain AUC 0.710 with leave-one-benchmark-out at chance; RouteLLM's released preference-trained routers score 0.523/0.533 zero-shot on our pool and rank the trap pool below ordinary hard pools; our router transfers to RouterBench below chance (0.440). On audio labels the router improves (test 0.814) but still trails the mid-layer probe (0.879) and cannot fire mid-stream. In distribution, the gate's area over random ($+0.054$) is statistically on par with the pool oracle's ($+0.042$; paired bootstrap delta CI $[-0.003, +0.027]$, p=0.13) — the gate's case is transfer, not the in-distribution margin. $p(\mathrm{True})$ tradeoff variants: Figure 9.

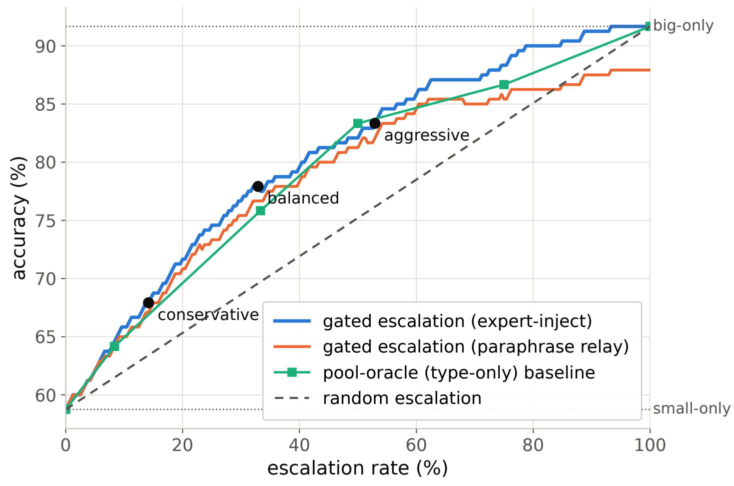


Figure 7: Accuracy vs. escalation rate, frozen test split: gated escalation vs. random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints.

F Gate timing details

Against the model's own thinking mode. MiniCPM-o 4.5 ships a built-in reasoning mode — self-escalation in place. Gated-cloud escalation dominates all-THINK on both accuracy and latency at every escalation rate (gated-cloud@33%: 78.7% at mean 6.5 s vs. all-THINK 63.7% at 22.4 s;

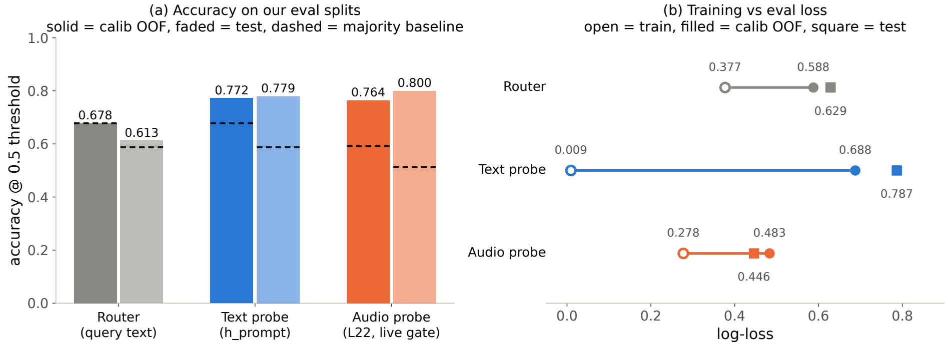


Figure 8: Identical training receipt for the external router and the two internal probes: the router never clears the majority-class baseline; both probes do, by 9–29 points.

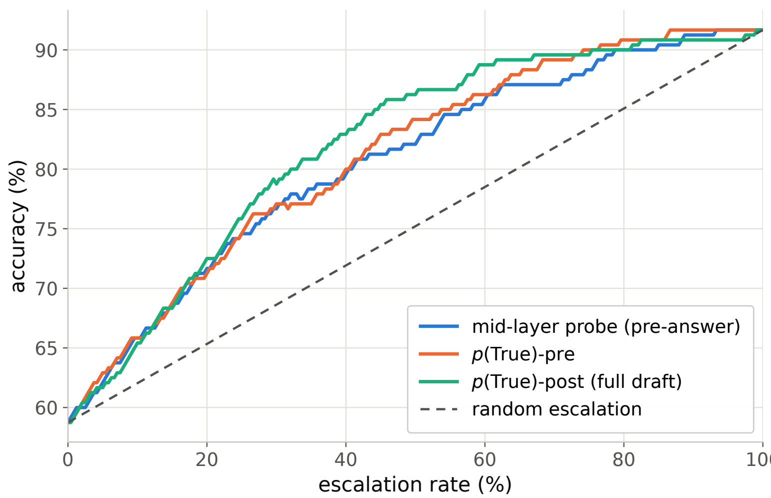


Figure 9: Offline tradeoff for the $p(\mathrm{True})$ signals (pre-answer and post-draft).

Table 9). The reason is mechanistic: the gate flags knowledge and trap failures, which extra compute cannot fix — so external escalation is necessary, not merely better.

Table 9: Deployment policies, frozen test split (n=240); latencies measured.

policy	acc (%)	lat. mean (s)	P50	P95
fast-only	58.8	3.0	3.5	4.2
all-THINK (built-in)	63.7	22.4	17.2	60.1
gated-think @33%	61.3	12.1	3.6	47.6
gated-cloud @15%	68.8	5.3	3.6	10.1
gated-cloud @33%	**78.7**	**6.5**	3.6	20.8
gated-cloud @50%	85.8	7.0	3.6	24.4

Measurement protocol. Gate latency is wall-clock from prefill start to the L22 score being available (truncated forward + probe dot product), CUDA-synchronized, one H100, 50 queries per modality, three warmup calls excluded, interleaved per query with the reference arms; P50/P95: gate 20/25 ms text and 45/104 ms audio vs. full-prefill TTFT 36/47 and 68/144 ms. Network and expert-RPC initiation are excluded: the cloud call fires asynchronously at the gate, and its user-facing cost is reported as expert-content-audible latency in Appendix G. Absolute prefill wall-times vary with call overhead across benches; the robust cross-bench statistic is the ratio — the L22 state is ready at ∼57–61% of prefill.

Per-layer truncated-forward cost vs. quality (Figure 10): the L22 hidden state is available at ∼57–61% of prefill wall-time while L22 is simultaneously the in-mix quality peak; forking much earlier is both weaker and modality-specific, so the fork belongs at ∼55–60% depth. If the cloud call launches at the gate, the expert result arrives before the local answer finishes for 40–58% of the whole mix but only ∼20–31% of *escalated* queries (they skew to short local answers); the residual wait is P50 2–3 s, P90 ∼27 s (Figure 11) — the stall-phrase budget the injection protocol must cover.

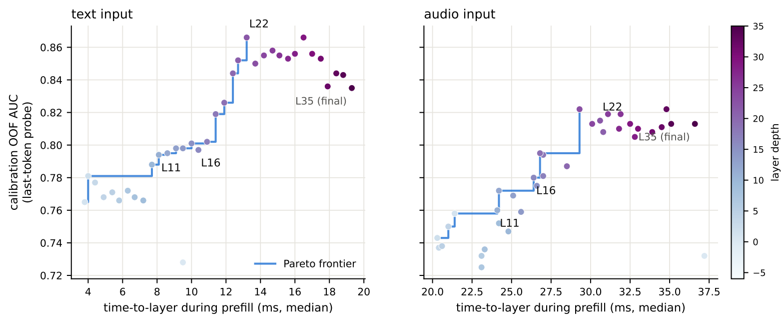


Figure 10: Fork-depth Pareto: per-layer wall-time vs. gate quality.

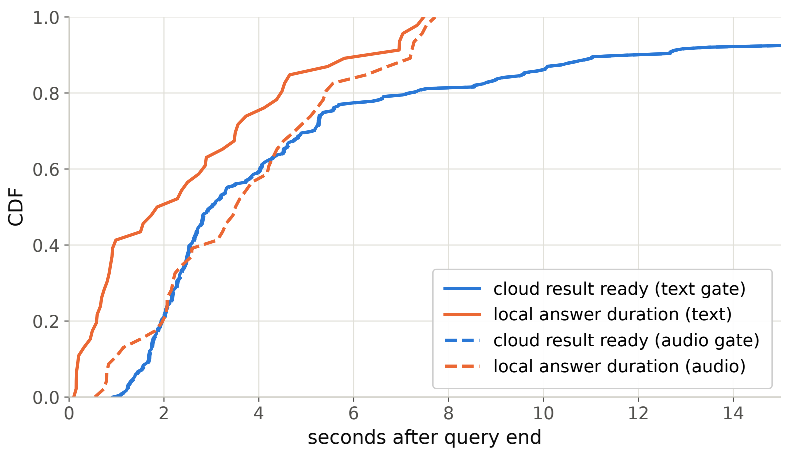


Figure 11: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends.

G Live-loop details

What the signal detects. The pre-answer probe is specifically a *retrieval-failure* detector, strongest on missing-knowledge and long-tail-trap failures where the empty lookup is observable in the model's state. Execution failures (stuck mid-derivation) surface in *decode-time* signals instead, and metacognitive failures — false-premise questions, where no failure event occurs at all — are invisible to every signal we tested, query-surface routers included (fire rate 18% vs. trap's 90%; Appendix G), which motivates a decode-time check behind the pre-answer gate.

Full live tables. Table 10 (both scoring views, speakable subset and full pool) and Table 11 (measured response latency). The conservative tier *shortens* the P95 tail: its few escalations displace exactly the longest local decodes. Figure 13 joins accuracy and latency; Figure 14 replays three real sessions from recorded timers.

Time to first audible audio. The sweep above is text-out; a TTS-on re-run of the balanced arm (same 240 queries, talker head + vocoder on, stall pre-synthesized once in the talker's own voice)

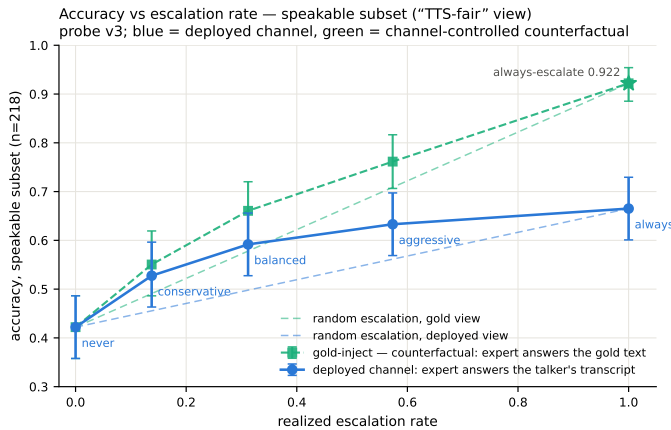


Figure 12: Live dual-view tradeoff (n=218 speakable subset). Blue: measured deployed-channel accuracy of the live system (paired-bootstrap CIs). Green: the channel-controlled counterfactual — same sessions, escalated rows re-scored with gold-text expert answers. Each view is plotted against *its own* matched-rate random reference (faint dashed, same floor, each view's own 100% endpoint): selection beats random within both views at every arm, by 0.072–0.095 deployed and 0.054–0.084 gold. The vertical view-to-view gap is the speech-channel price; the floor-to-floor gap to offline operation is the audio + streaming-answer tax.

measures speech onset directly. The gate decision is unchanged — the TTS token enters the context only after the end-of-turn read (read p50 21 ms / p99 32 ms), and the same 84/240 queries fire. From gate decision to first audible audio: **local answers p50 0.552 s / p95 0.599 / p99 0.644**; **escalated turns p50 0.022 s / p99 0.029** — the canned stall (3.2–4.1 s of speech) plays the moment the gate fires, so the escalated path reaches first audio 25× faster. Expert *content* becomes audible at p50 6.4 s / p95 26.0 / p99 34.0 (cache-corrected expert round-trip + first relay PCM, p50 0.66 s after the answer returns); per-turn dead air between stall end and expert content is p50 2.5 s / p95 22.4 / p99 30.7, with 25/84 escalated turns fully covered — the stall covers the head of that wait, and Table 11's completion profile bounds the tail.

Table 10: Live streaming sweep, paired bootstrap 95% CIs (B=10^4). Top: speakable subset (n=218); bottom: full pool (n=240). Gold-inject = escalated rows re-scored with gold-text expert answers; channel cost = paired difference. * CI excludes 0; † CI lower bound at zero.

tier	esc	deployed	Δ vs. floor	gold-inject	channel cost
never	0%	0.422 [0.36,0.49]	—	0.422	—
conservative	14%	0.528 [0.46,0.59]	+0.106*	0.550	+0.023†
balanced	31%	0.592 [0.53,0.66]	+0.170*	0.661	+0.069*
aggressive	57%	0.633 [0.57,0.70]	+0.211*	0.761	+0.128*
always (measured)	100%	0.665 [0.60,0.73]	+0.243	0.922 [0.89,0.95]	+0.257
never (full)	0%	0.383 [0.32,0.45]	—	0.383	—
conservative	16%	0.483 [0.42,0.55]	+0.100*	0.525	+0.042*
balanced	35%	0.554 [0.49,0.62]	+0.171*	0.654	+0.100*
aggressive	61%	0.596 [0.53,0.66]	+0.212*	0.771	+0.175*
always (measured)	100%	0.617	+0.234	0.917 [0.88,0.95]	+0.300

Speculative prefetch, measured and dropped. Framed as speculative escalation (draft–verify at sentence granularity), its acceptance rate — does an expert answer to the partial transcript answer the full question — is 0.07/0.19/0.51 at 25/50/75% of the audio; at the mid-stream gate's typical fire point 6–8 of 10 calls would be wasted. The trap pool's floor (0.00/0.10/0.27) shares the back-loaded-core property that breaks mid-stream probe reads.

Table 11: Measured response latency (s), n=240 per arm (latency is channel-independent). P99 on n=240 is indicative only.

tier	mean	P50	P95	P99	Δmean	ΔP50	ΔP95	ΔP99
never	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

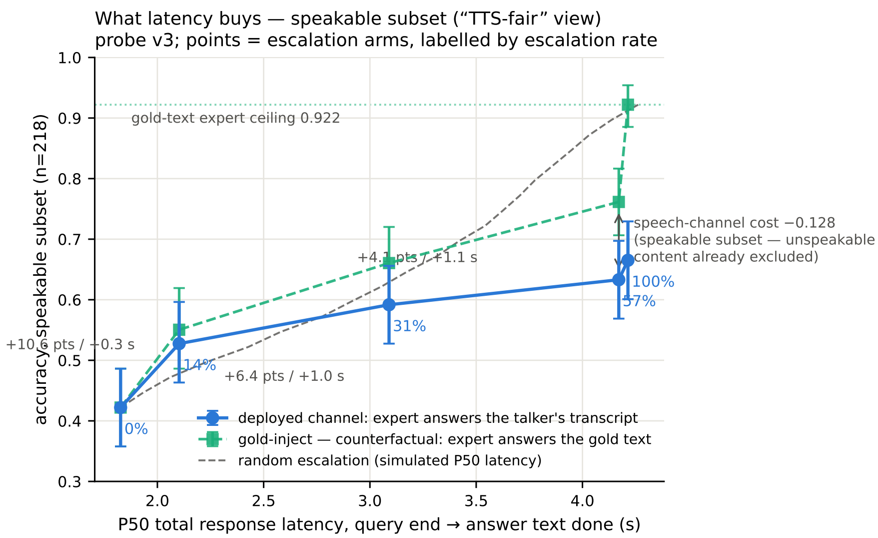


Figure 13: Accuracy vs. per-arm P50 latency (speakable subset), both scoring views, with the simulated-P50 random reference.

Conflicting-injection controls (172 sessions). Fabricated conflicting expert answers by within-pool derangement, four framings: worst-case comply is 0.00 neutral / 0.53 deployed authority / 0.79 strong-override, while *seeding* words into the talker's mouth backfires (comply 0.25, lip-service 0.62 — forced speech is treated as revisable). Math is silently re-derived rather than relayed or disputed. Model-wrong $\times$ wrong-injection swallow rate is 0.64: expert quality is the accuracy ceiling of the cascade.

Uplink diagnostics. On the escalated population the expert answering the talker's own transcript scores 0.585 vs. 0.871 on gold text. Prompt-level rescues on the self-transcript are no-ops (the damaging errors are internally-coherent entity substitutions); replacing the ears helps: open-weights Whisper recovers little (+4 pp) but a stronger cloud ASR on the same audio recovers 38% of the gap (0.585$\rightarrow$0.694, McNemar p=0.007) — diagnostic only (its critical-path latency and external-pool survival are unmeasured). A same-model gold-text control with an audio-capable expert shows the audio channel itself is lossless for speakable content: the leak is the talker's transcription, not the synthesis or the relay (relay drops: 5 cases of 108; corrupted transcripts dominate the escalated failures, first-generation sweep).

Boundary query classes. *False premises*: the model fails substantially (0.63 text / 0.47 audio) and the expert would help (0.80), but the gate is blind (fire@balanced 0.18 vs. trap's 0.90; no score separation) — answering along a false premise does not feel like inability. *Real-time queries* ("what is NVDA trading at"): the gate is not blind, the class was simply absent from every static training family; extending calibration with FreshQA [Vu et al., 2023] (fast-changing labeled escalate a priori, never-changing as in-family controls) raises held-out fast-changing fire rates to 0.80/0.93/1.0 across tiers with frozen-test AUC unchanged (0.877 vs. 0.879). One calibration subtlety: tier budgets must

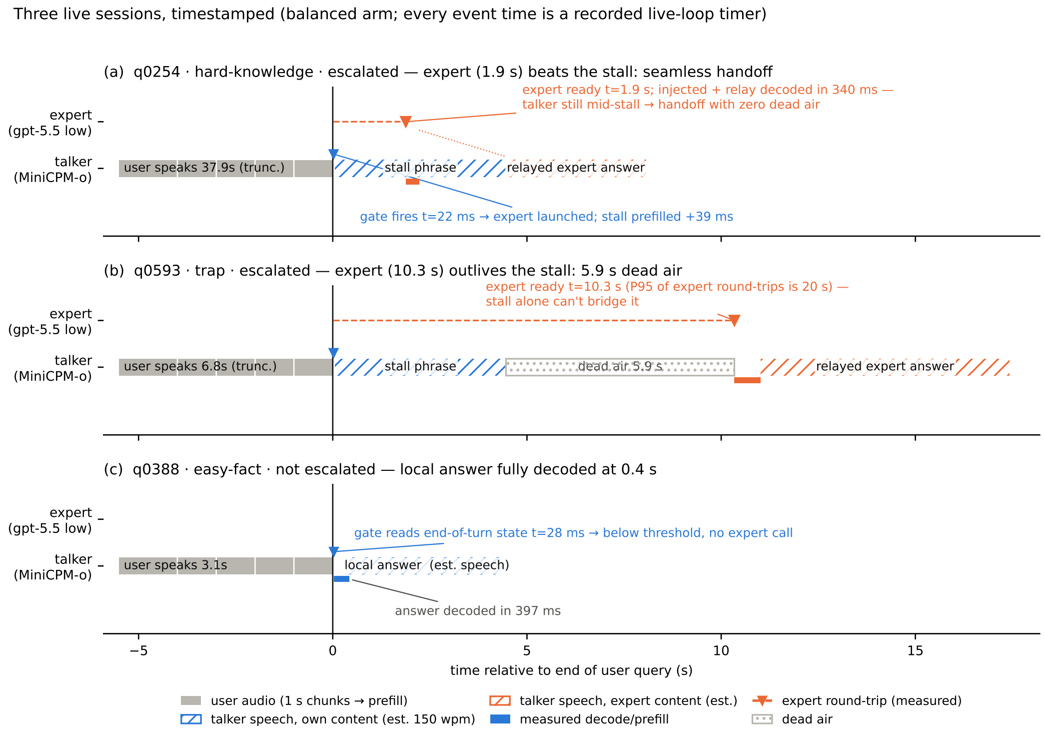


Figure 14: Three real sessions from recorded timers: expert beats the stall phrase (seamless); expert outlives it (dead air, the P99 signature); gate stays closed (local answer fully decoded 0.4 s after end of turn). Speech occupancy is estimated at 150 wpm (hatched); the measured arms are text-out, so no vocoder onset is plotted.

remain quantiles of the deployment-mix scores, or the new capability is spent on its own threshold shift.

Demonstration system. The pipeline runs as an interactive web demo on one H100, on the model's *native full-duplex* head (Appendix O): every second of microphone audio is prefilled into the context the talker is generating from, the head itself decides listen/speak per chunk, and the gate reads L22 at the listen→speak transition against the in-regime thresholds. Escalated turns play a canned stall (teacher-forced once, the talker's own voice), consult the expert (with a demo-only web-search tool for real-time queries), and the verified answer re-enters the stream as a text unit that the talker voices — an ordinary duplex turn, hence interruptible like any other. There is no VAD, no ASR barge-in classifier, and no abort machinery: floor control is entirely the duplex head's own behavior (Appendix O). An earlier demo build approximated barge-in with an energy-VAD + burst-ASR harness over the turn-based loop; it is retired, and survives only as the harness-overlap arm measured above. The demo will be made available upon publication.

H Duplex-validation sweep: the gate under the talker

The headless live sweep (§6) never engages the talker head; this sweep replays its full 240-query pool through the deployed voice loop above to test whether the frozen probe read survives the spoken regime. Two arms, one session per query, deployed v4 gate with scores logged but escalation disabled (the arm isolates gate validity, not routing outcomes). **Clean**: the query streams as mic-shaped PCM (pool TTS audio plus steady noise at RMS 0.008), the energy VAD ends the turn, the talker answers aloud, and the session is cut after the first audible chunk. **Overlap**: an easy warmup question (probe score $<$0.25, locally answered, $>$5 s spoken answer) puts the talker mid-answer, and the target query is then spoken over it; sustained overlapping speech commits a barge-in whose audio seeds the next turn. The barge-in path engaged on 237/239 pairs.

Against the frozen local-failure labels of the same pool, the identical frozen probe reads: headless AUC 0.866 $[0.817, 0.909]$, clean 0.871 $[0.824, 0.914]$, overlap 0.856 $[0.803, 0.901]$; barged-only subset (n=237) 0.854. Paired bootstrap deltas vs. headless are $+0.005$ $[-0.025, +0.035]$ (clean) and -0.009 $[-0.036, +0.016]$ (overlap) — both inside the ±0.02–0.03 live replication floor. Per-query scores correlate across regimes at r=0.90–0.92, and the balanced threshold makes the same fire/hold decision as the headless read on 0.887 (clean) and 0.912 (overlap) of queries. The end-of-turn read costs 24–26 ms at p50 (p95 $\leq$30 ms) in both arms, with the first audible answer chunk arriving $\geq$0.5 s after the gate decision (p50 $\approx$0.58 s), so the decision precedes any audible commitment. Scope: this loop keeps per-turn session control — the mic stays open while the talker speaks and interruptions seed the next read, but incoming audio is not prefilled *during* generation. In hindsight this sweep and the concurrent arm are best read as *read-point ablations*: the deployed native full-duplex regime (Appendix O) sits strictly between the turn-based ideal and the harness-concurrent worst case on every pool measured. The concurrent regime is measured separately below.

Floor control under escalation: barge-in vs. backchannel. The demonstration loop's floor policy (speech over the talker ducks playback; a fast ASR pass returns the floor on backchannels or commits the interrupt) raises a distinct worry: did adding the gate change the talker's overlap behavior? A dedicated sweep injects overlap stimuli into the deployed loop at real-time pacing — four classes: short backchannels (0.40–0.46 s, the English+Chinese lexicon), long lexicon-only continuers (time-stretched to 1.33–2.08 s, probing the sustained-speech commit rule), out-of-lexicon stop commands, and pool queries spoken over the answer — in matched pairs with the gate on and off under an identical floor policy (non-firing queries, n=40 per class per arm). The answer is exact orthogonality: all 160 pairs make the identical duck/resume/interrupt decision in both arms (every paired rate delta 0.000 $[0.000, 0.000]$; paired latency medians within 31 ms), down to the single seeding failure, which occurs on the same query in both arms. By construction the gate reads once at end of turn and never enters the floor state machine; this measures it. The three phases that exist only under escalation — the stall sentence, the expert wait, the relay speech — are the actual new risk surface and behave (Table 12): a genuine barge-in aborts the escalated turn and seeds the next turn 40/40, and short backchannels hold the floor at the same rate as during local answers. One asymmetry is disclosed: a false interrupt during the expert wait cancels the pending expert call, so the policy's fail-toward-interrupt bias is recoverable but not free there. That residual backchannel false-interrupt rate (0.40 short, 0.50 long; identical in both arms) is a property of the harness policy and is per-stimulus deterministic: non-lexical hums ("mm-hm" and its Chinese counterpart) die by the ASR-empty fail-closed rule (16/16 each; worded tokens such as "okay" 0/16), and continuers sustaining loud speech past the $\geq$1.2 s commit rule die without any ASR check (20/20; pause-bearing variants 0/20). We keep the policy as deployed for measurement integrity and note both rules as the obvious fixes. For scope: the checkpoint's *native* duplex profile, measured separately on Full-Duplex-Bench v1.0 [Lin et al., 2025] (727 samples, official scripts), shows best-in-class pause handling (turn-over rate 0.125/0.117 Candor/synthetic), competitive user-interruption response (0.915 at 0.90 s), and *zero* spontaneous backchannels — the lexicon floor is a harness-level approximation added precisely because the talker head has no native backchannel behavior to preserve.

Concurrent prefill: interleaved listen/speak in one KV stream. A third arm drives the streaming API in time division: while the talker is mid-answer, each 1 s chunk of the target query is prefilled into the *same* session between generation chunks, so the end-of-turn read fires while generation is still active (238/240 queries, full interleave coverage). The carrier answer is teacher-forced: under sampling the duplex model reliably yields the turn — it emits EOS within $\sim$2 s of user audio appearing in its context, the trained turn-taking behavior surfacing under interleave. This is the one shift on the transfer ladder that breaks the frozen read: scores drift up (mean 0.48$\rightarrow$0.71; the frozen balanced threshold's fire rate jumps 37$\rightarrow$75%), AUC drops 0.869$\rightarrow$0.758 $[0.693, 0.816]$ (paired ΔAUC -0.109 $[-0.158, -0.064]$), and label-free requantiling recovers decision agreement only to 75%. The information, however, survives in the state: refitting the *same* 12288-d linear read on 360

overlap injected during	backchannel holds floor	barge-in aborts	seeds next turn
local answer (gate off)	0.60 [0.45, 0.75]	40/40	39/40
local answer (gate on)	0.60 [0.45, 0.75]	40/40	39/40
stall sentence	0.63 [0.38, 0.88]	16/16	16/16
expert wait	0.60 [0.33, 0.87]	14/14	14/14
relay speech	0.55 [0.27, 0.82]	10/10	10/10

Table 12: Floor-control sweep over the deployed voice loop. Short backchannels (lexicon bursts) vs. genuine barge-ins (pool queries spoken over the talker), by the phase the overlap lands in; the last three phases exist only under escalation. Interrupt latency p50 1.39–1.42 s from overlap onset in every phase; escalated-turn aborts discard the pending expert answer and the barge-in speech becomes the next turn.

calibration queries collected in-regime reaches **0.818** [0.761, 0.871] on the held-out 240 (calibration OOF 0.775) — more than half the gap recovered at a quarter of the deployed probe's calibration scale. The mid-layer competence direction persists under concurrent listen/speak; unlike every earlier shift (modality, external pools, talker-on), calibration must follow the regime. Run-to-run replication of the concurrent read correlates at 0.895. Caveat: the interleave goes through the public streaming API — turn-role headers enter mid-generation and the carrier text is forced — so this measures the concurrent *state*, not the official duplex serving format.

The full loop in the concurrent regime. With the in-regime gate (thresholds = label-free quantiles of each pool's own concurrent never-arm scores), the complete escalation loop — carrier speaking, target audio interleaved, EOT read mid-generation, stall $\rightarrow$ expert $\rightarrow$ relay — was re-run end to end on all seven pools (Table 13). We report this sweep as *exploratory*: one session per query per arm, an in-regime probe calibrated on 360 internal rows only, teacher-forced carrier (the caveat above). The loop's mechanics reproduce — realized escalation rates track nominal on the external pools (10.8–51.2%), and latency holds (EOT read p50 23 ms, canned-stall onset 27–34 ms, local first audio p50 .655 s vs. .552 turn-based, the price of the longer concurrent context) — and on the internal pool, where calibration matches the regime, so does *selection*: accuracy climbs 40.4$\rightarrow$52.1$\rightarrow$66.2% (speakable **44.5$\rightarrow$56.0$\rightarrow$71.1%**) over the balanced and aggressive tiers, clearing the matched-rate random remix of the same never/always outcomes (p=.017 and 5×10^{-5}; speakable .009 and 5×10^{-5}), and *selective escalation beats always-escalate* (66.2 vs. 61.7% at two thirds of the calls; speakable 71.1 vs. 66.5) — the turn-based signature result carries over. The matched-random rows of Table 13 bound any stronger reading, in three ways. (i) Gains are not monotone everywhere: the internal conservative tier under-fires (2.5% realized vs. 15.8% nominal — the top calibration quantile did not transfer to the regime; balanced and aggressive did) and lands flat (40.4$\rightarrow$40.0, 44.5$\rightarrow$43.6, inside the $\pm$2–3 floor). (ii) Externally the 360-row in-regime probe separates from matched-rate random only in spots (Llama @50%: 90.8 vs. 86.5, p=3×10^{-4}; Reasoning zh @30/50%: $p \leq .0024$; AlpacaEval @15%: p=.031); at the other external operating points the budget gains are what random escalation would buy — consistent with this probe's external AUC of .57–.67 against the turn-based probe's .68–.81 (internally, where calibration matches the distribution, it reads .857 vs. the turn-based .879). We then tested the calibration-scale attribution directly: the full 2,310-row calibration mix was re-collected in the concurrent state and the same read refit at full scale. Scale is real — every English pool lifts (paired against the 360-row probe on the same features: TriviaQA .645$\rightarrow$.699, WebQ .641$\rightarrow$.713, Llama Q. .674$\rightarrow$.755, SD-QA .639$\rightarrow$.701; mean ΔAUC +.068 [.036, .099]) along a monotone data-scaling curve (external mean .625$\rightarrow$.689 over 360$\rightarrow$2,310 rows; internal .818 unchanged) — but it buys only about a third of the distance to the turn-based .785–.806, and Reasoning zh, absent from the English calibration mix, moves within noise (.615$\rightarrow$.577 [$-$.117, .037]). The residue is the regime itself, not calibration scale. Table 13's loop ran with the 360-row probe and its numbers are unchanged. (iii) On Llama Questions selective-vs-always is judge- and noise-sensitive (official judge 90.8 vs. 91.2; ours 86.4 vs. 85.6 — both deltas inside the floor), so we claim that comparison only on the internal pool. Floors move by pool — the regime cost story: Llama $\approx$ unchanged, TriviaQA $-$16, Reasoning zh $-$18 (the English carrier hurts the zh pool most), WebQ +4.8 under the official judge (within the ±2–3 replication floor; the larger our-judge shift is alias-matching style sensitivity). AlpacaEval rises 4.36→4.60 (1–5 scale, always 4.76), but only its conservative tier clears matched random (p=.031; 4.40–4.55 across tiers), so the turn-based honest-negative stands.

Table 13: The full escalation loop in the concurrent regime (*exploratory*: one session per arm, 360-row in-regime calibration, teacher-forced carrier), all pools, judge-matched to Table 1 (OAB official for TriviaQA/WebQ/Llama; ours for SD-QA/Reasoning/internal; VoiceBench 1–5 for AlpacaEval). Gate = in-regime probe, per-pool label-free quantile thresholds; tier columns are headed by each pool's *realized* rate, and the *rand* row under each pool is the matched-rate random remix of the same never/always outcomes (20k draws; p = P(random ≥ gate), marked $^*p<.05$, $^{**}p<.01$). Accuracies in %; AUC = in-regime probe vs. never-arm failure, decimals. Script: `scripts/20_conclive_random_check.py`.

pool	never	cons.	bal.	aggr.	always	AUC
TriviaQA	50.4	54.0	64.0	71.6	91.6	0.636
rand (10.8/30.8/46.8%)		54.9	63.1	69.6		
WebQ	61.6	65.2	68.8	71.2	81.6	0.636
rand (15.6/30.4/48.8%)		64.7	67.7	71.4		
Llama Q.	81.6	83.2	84.8	90.8**	91.2	0.672
rand (14.8/29.2/51.2%)		83.0	84.4	86.5		
SD-QA	49.5	55.5	60.5	70.5	89.5	0.636
rand (15.0/29.5/50.5%)		55.5	61.3	69.7		
Reason. zh	40.6	48.5	58.4**	64.4**	80.7	0.570
rand (15.3/30.2/45.0%)		46.8	52.7	58.7		
internal (240)	40.4	40.0	52.1*	66.2**	61.7	—
rand (2.5/36.2/66.2%)		41.0	48.1	54.5		
internal speakable (218)	44.5	43.6	56.0**	**71.1****	66.5	0.857
rand (2.3/31.2/62.8%)		45.0	51.4	58.3		
AlpacaEval (1–5)	4.36	4.45*	4.53	4.60	4.76	—
rand (10.1/30.7/47.7%)		4.40	4.48	4.55		

A serendipitous recovery behavior. An accidental run surfaced one bonus behavior worth reporting: when a long question contains a >1.25 s internal pause, the endpointer fires early and the talker starts answering the fragment — but the question's own continuation then triggers barge-in, and the remainder becomes the next turn, where the gate simply re-scores it. Early end-pointing thus degrades into an extra turn rather than a lost query — the honest duplex-ish recovery available to a turn-based loop.

I Fixed-threshold transfer and the expert-benefit oracle

Both analyses below run in the *gold-inject* view (escalated rows take the expert's answer to the gold text). The deployed-channel view cannot evaluate a freely moving threshold: the most permissive arm escalates 50%, so the other half of each pool has no expert outcome on record, whereas the ceiling runs cover every id. Labels follow each pool's own judge (Table 1); the two judges disagree by up to 13 points on Web Questions, so they are never mixed within a row. AUC columns are recomputed from the deployed sweep's stored end-of-turn scores; they differ from Table 1's AUC column — captured in a separate scoring pass — by up to 0.03 externally and 0.04 on the internal pool (0.840 here vs. 0.879 there): the passes render and prefill the audio independently, and the probe score carries render-to-render jitter. AlpacaEval is excluded — it has no binary correctness label on either side, only a 1–5 score whose expert ceiling is degenerate (195/199 at 5).

One absolute threshold, six pools. The deployed gate adapts per pool, but only through label-free quantiles of its own score distribution. To separate what the probe's *ranking* carries from what its *scale* carries, we freeze the single absolute threshold the internal system actually deployed (τ=0.628, the balanced tier) and apply that number verbatim everywhere. **Scale does not transfer**: the same τ realizes escalation rates from 2.0% (Reasoning QA) to 48.0% (SD-QA), a 24-fold spread, so a fixed cut cannot hold an escalation budget across pools. **Ranking does**: at whatever rate it fires, fixed-τ escalation beats a random reference at the same realized rate on four of five external pools, paired-bootstrap interval excluding zero (Table 14). The exception is Reasoning QA, where τ fires on 2% of queries and there is almost nothing to measure. This is the concrete case for the label-free quantile step: it buys budget control, not discrimination.

Does the gate find failures the expert can fix? The probe is fit and read against *local failure*, but what a routing system needs is *expert benefit*: $y{=}1$ when the local model is wrong *and* the expert is right. Re-scoring the same frozen scores against this harder label costs little on the retrieval pools (Speech TriviaQA 0.796→0.782, Llama Questions 0.830→0.813) and more where the expert is itself weak (Web Questions 0.773→0.689 against a 0.864 expert ceiling; internal 0.840→0.732). The gate thus remains a usable benefit predictor out of distribution (0.63–0.81) despite never having seen an expert outcome, and the drop is largest exactly where the paper already reports a flat live curve. On Web Questions the mechanism is direct: 75.0% of local failures are expert-fixable pool-wide, but only 65.5% of the failures inside the gate's top-scoring 15% are, and that share climbs monotonically as the cut widens (0.655/0.685/0.711 at 15/30/50%) — the gate's most confident failure calls are the ones the expert also misses. A benefit-*trained* refit closes none of this: with expert outcomes collected for all 2,310 calibration queries (gold-text expert, expert-right rate .853), retraining the same 12,288-d read directly on the benefit label moves internal benefit-AUC .732→.759 (paired $+.027$ [.004, .052]) and leaves every external pool inside noise (Δ $-.029$ to $+.037$, all CIs spanning zero), at no internal cost on the fail label (.840→.839). The benefit gap reflects the label's dependence on the expert's own failure surface, not a trainable objective mismatch — the expert-agnostic label leaves nothing on the table. Table 15 places the deployed gate against the matched-budget oracle that escalates true benefit cases first: the gate clears random at every pool and budget, capturing a fifth to three fifths of the random-to-oracle span, and the residual gap is the headroom a decode-time or expert-aware second stage would have to claim.

Table 14: One absolute threshold ($\tau{=}0.628$, frozen on the internal pool) applied verbatim to every pool; gold-inject view. Δ is fixed-τ accuracy minus a random reference at the *same realized rate* (paired bootstrap over ids, $B{=}10^4$, seed 42). AUC columns re-score the same frozen scores against local failure and against expert benefit ($y{=}$ local wrong $\wedge$ expert right, base rate in the last column). Rates and accuracies in %; AUC in decimals.

pool	n	rate	local	acc	rand	Δ 95% CI	AUC$_{\text{fail}}$	AUC$_{\text{benefit}}$	base
Speech TriviaQA	250	28.4	66.4	83.2	75.0	$[+4.4, +12.1]$	0.796	0.782	30.8
Speech Web Questions	250	36.0	56.8	72.0	67.5	$[+0.8, +8.4]$	0.773	0.689	32.4
Llama Questions	250	8.0	83.6	86.8	84.3	$[+0.6, +4.6]$	0.830	0.813	11.2
SD-QA	200	48.0	51.5	80.5	71.4	$[+4.5, +13.6]$	0.800	0.765	43.5
Reasoning QA (zh)	202	2.0	58.9	60.9	59.5	$[-0.1, +3.5]$	0.679	0.634	30.2
internal frozen test	240	35.0	38.3	65.4	57.0	$[+3.8, +13.1]$	0.840	0.732	53.3

Table 15: Oracle frontier at matched escalation budgets (gold-inject view): random reference / deployed quantile gate / benefit oracle, which escalates the true $y{=}1$ cases first. The gate clears random everywhere; the oracle column is the remaining headroom. Accuracies in %.

	15% budget			30% budget			50% budget		
pool	rand	gate	oracle	rand	gate	oracle	rand	gate	oracle
Speech TriviaQA	71.0	76.0	81.6	75.5	83.6	96.4	81.6	91.2	97.2
Speech Web Questions	61.2	64.0	72.0	65.7	70.0	86.8	71.6	76.4	89.2
Llama Questions	85.0	88.8	94.8	86.4	91.2	94.8	88.2	93.2	94.8
SD-QA	57.7	63.5	66.5	63.9	74.5	81.5	72.2	82.0	95.0
Reasoning QA (zh)	63.1	67.8	73.8	67.4	72.8	89.1	73.0	76.7	89.1
internal frozen test	46.3	49.2	53.3	54.3	60.8	68.3	65.0	74.2	88.3

J Transfer latency shapes

Matched-rate precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail 0.16) the 50% cap is 0.32 and the deployed probe reaches 0.304 — 95% of the cap (recall 0.93). Quoting raw precision would misread the strongest pool as the weakest.

The median left-fold. On easy pools per-arm P50 is non-monotonic (Llama Questions 1.52→1.17→1.60/1.42 s): a verified median-of-a-mixture mechanism, not noise. The probe preferentially escalates the queries whose *local* decode is longest — on retrieval pools wrong answers hedge at length (decode time and answer length r=0.97), so a wrongness-selecting probe strips the slow tail as a side effect (escalated-at-15% ids: never-arm local P50 1.711 s vs. 1.198 s kept; local accuracy 0.32 vs. 0.73). Controls: the probe does not key on length (corr(score, length) = +0.05; ≈0 conditioned on wrongness), and on SD-QA — where wrong answers are short — the same probe selects wrongness identically and produces no fold. The mean is monotonic (1.65→2.06 s); kept-local P50 falls 1.52→0.43 s across tiers (Figure 15). On Reasoning QA the shape inverts usefully: the local chain-of-thought is *spoken*, so every reasoning token enters the response clock, while the expert's chain is hidden behind a flat ∼3.7 s round-trip — escalation truncates the latency tail (P90 13.25→10.94 s) rather than fattening it.

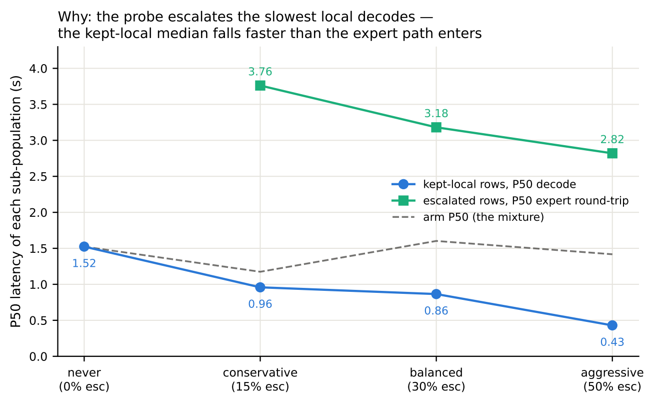


Figure 15: Mixture decomposition of Llama Questions latency per arm: kept-local P50 falls 1.52→0.43 s as the probe strips the slowest local decodes while escalated rows pay a flat ∼3 s expert round-trip; the arm median (dashed) is their mixture.

AlpacaEval scale caveats. The official 4.8 is an offline chat-mode number — our chat-mode control reproduces it at 4.86 — and the chat-vs-live gap (4.86 vs. 3.99) is the duplex loop's spoken-reply style plus a token cap, so only live arms are comparable to each other. Selected queries score 3.90 on the never arm vs. 3.98 for skipped: no discrimination; escalated rows gain (3.90→4.71) because the expert writes better long-form answers — random picks would harvest the same gain.

K Full-pool and per-family figures

Figures 16 (dual-view) on the full frozen pool, and the complete external-family figure set (Appendix L); the noise audit behind the ±0.02–0.03 floor is Figure 17.

L External-benchmark figure set

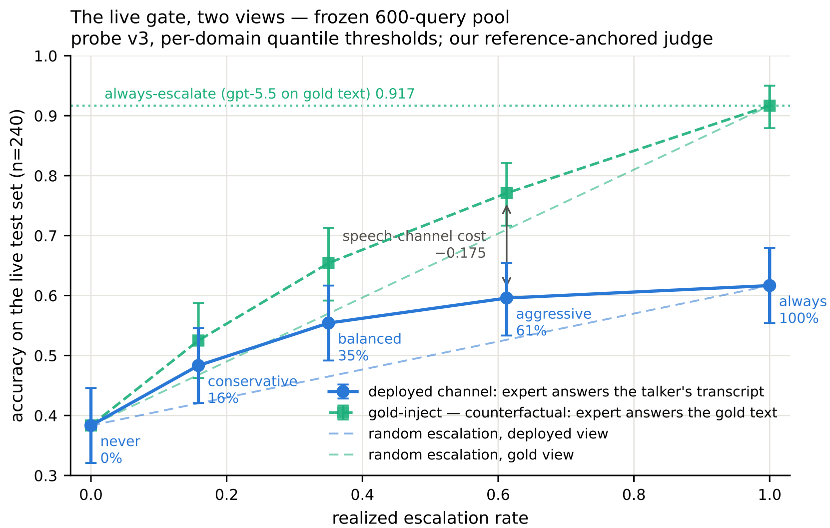


Figure 16: Dual-view live tradeoff, full pool (n=240); companion to Figure 12.

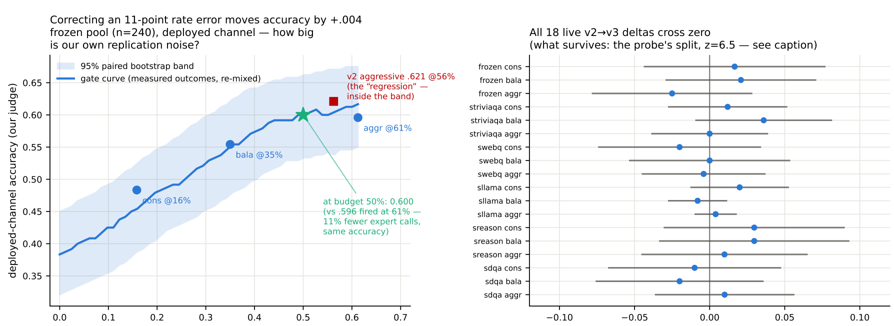


Figure 17: Replication-noise audit: same query, same audio, re-run across arms. Kept-local verdicts flip 2.3–18.8% by pool; repeated escalation 0.7–10.6%; implied paired SE 0.009–0.028.

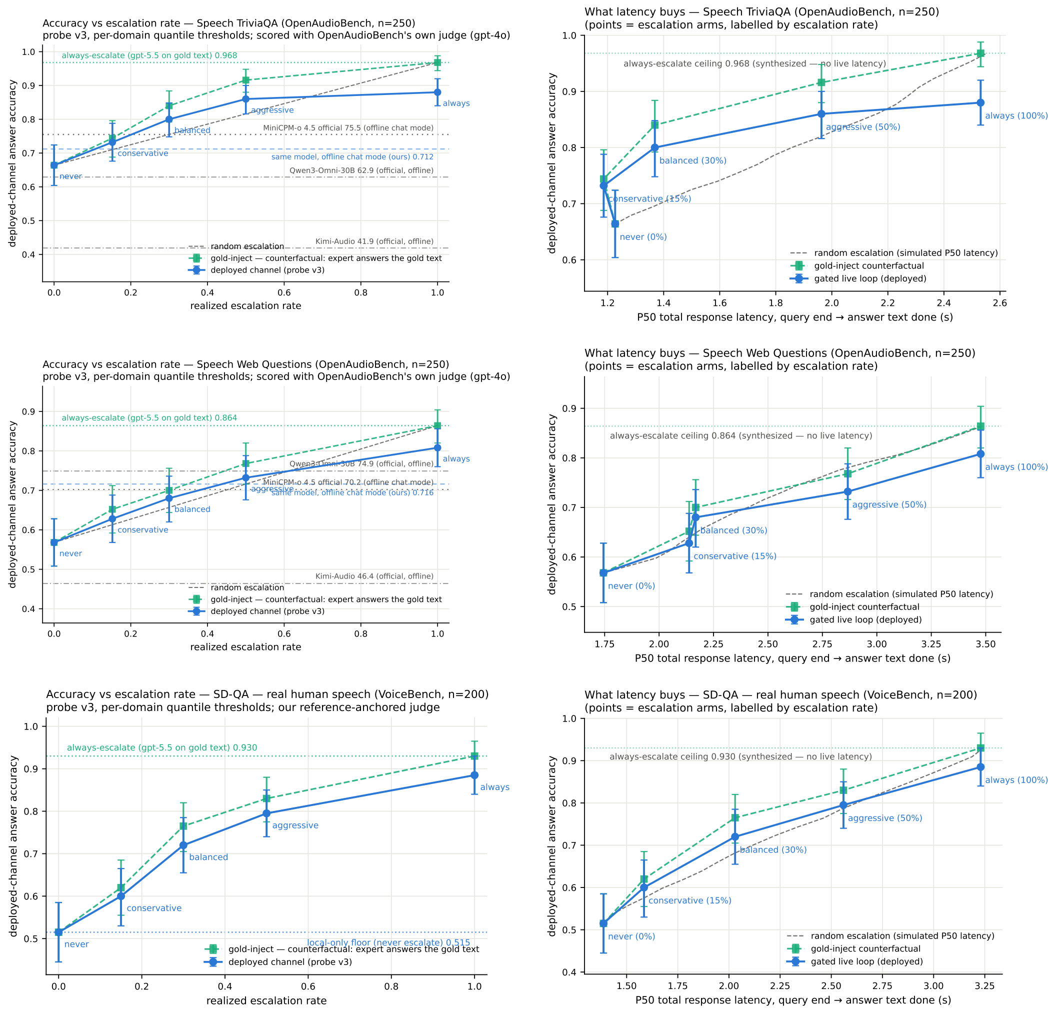


Figure 18: Speech TriviaQA, Speech Web Questions, SD-QA: dual-view (left) and accuracy–latency (right).

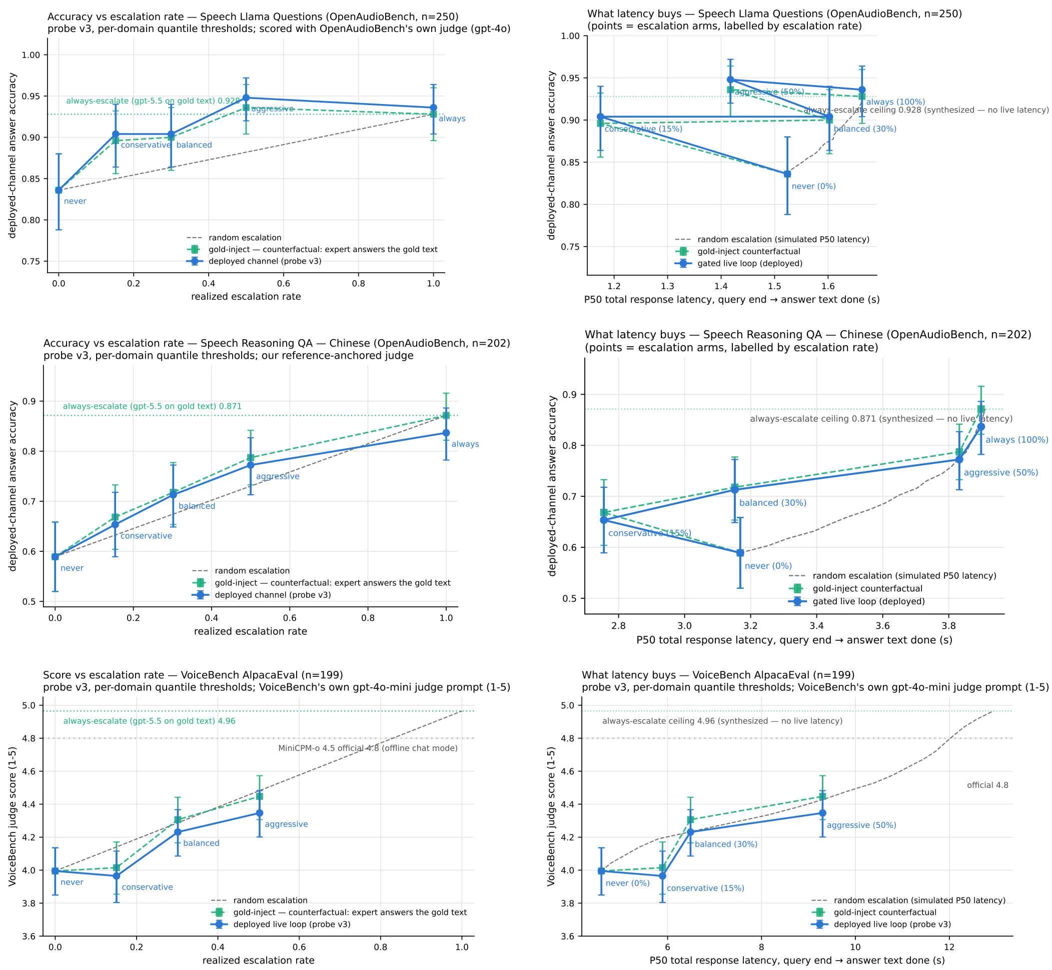


Figure 19: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: dual-view (left) and accuracy–latency (right).

M Judge and self-evaluation prompts

All adequacy labels use the following fixed system prompt (structured outputs with a strict {`reason`, `adequate`} schema; the judge never sees which system produced the answer):

```
You are a strict, fair grader of a small language model's answers.
You are given a QUESTION, an optional REFERENCE answer, and a
CANDIDATE answer produced by a small model.  Decide whether the
CANDIDATE is ADEQUATE, meaning ALL of:  (1) factually correct;
(2) any reasoning shown is valid; (3) it actually and completely
answers the question that was asked.  For multiple-choice questions
the candidate must land on the correct option (by letter or by
unambiguously stating the option's content).  A partially correct,
hedged, refused, off-topic, or truncated answer is NOT adequate.  If
a REFERENCE is provided, grade against it.  Be strict but do not
penalize harmless differences in format or extra correct detail.
```

The verbalized self-evaluation baselines use one fixed phrasing each; $P(\text{Yes})$ is read from the first generated token's distribution:

```
[pre] I am going to show you a question.  Do NOT answer it.  Judge
honestly whether you yourself would answer it correctly.  Question:
{q} Would you answer this question correctly?  Reply with exactly one
word:  Yes or No.
[post] Here is a question and a proposed answer.  Question:  {q}
Proposed answer:  {a} Is the proposed answer correct?  Reply with
exactly one word:  Yes or No.
```

N Reproducibility

All sampling and splits use a fixed seed (42); the calibration/test split is frozen once and the test split is touched only by the evaluation scripts. Environment pins: torch 2.8.0; transformers 4.51.0 for MiniCPM-o 4.5, with per-model pins for comparison backbones; all experiments run on single-H100 instances. Probe artifacts (weights, feature configuration, per-pool label-free thresholds), the judge prompt above, all trace files, and the per-experiment execution log — including per-run GPU and API spend, on the order of \$500 total — ship in the project repository, released upon publication.

O The native full-duplex regime

Every result so far reads the probe under a *turn-based* serving loop (harness-controlled end-of-turn) or its harness-interleaved concurrent variant (Appendix H). MiniCPM-o 4.5 also ships a native full-duplex mode: audio prefills in 1 s units into the same context the model generates from, and the head itself emits $\langle$listen$\rangle$ or speaks, yielding the floor via its turn-end token. This is the deployed regime of the demonstration system, and this section recalibrates and re-validates the gate inside it. The read point needs no external end-of-turn detector: the chunk where the head first switches listen$\rightarrow$speak *is* the commit-to-answer moment, and the probe reads there (the tail-k features then cover the model's first $\sim$8 answer tokens — a generation-side read the head hands us for free). For 87% of test queries this commit lands within $\pm$1 chunk ($\pm$1 s) of the true question end.

Recalibration. The full 2,310-row calibration mix was re-collected in this regime (fresh duplex session per query, features at the deployed read point, labels unchanged) and the same 12,288-d linear read refit. On native test-240 features the in-regime probe reaches AUC **0.830** [0.777, 0.878] (transferred concurrent probes: 0.807–0.818; turn-based reference 0.877). Externally the in-regime refit beats both transferred probes on all five pools — TriviaQA .711 (Δ+.080 [+.030,+.129]), WebQ .736, Llama-QA .757 (Δ+.118), SD-QA .736, Reasoning-zh .606 (Δ+.115); external mean .709. The regime ordering is monotone everywhere: turn-based (.877/.771 internal/external mean) > native (.830/.709) > concurrent (.689 external) — i.e. a substantial share of the concurrent regime's cost

Table 16: Native-regime probe AUC (features at the deployed listen→speak read point). Transferred concurrent probes vs. the in-regime refit; paired bootstrap Δ vs. the 2,310-row concurrent probe. The in-regime refit wins on every pool.

pool	n	conc-360	conc-2310	**native**	Δ(nat−2310)
internal test	240	.818	.807	**.830**	+.023 [−.012,+.059]
TriviaQA	250	.658	.631	**.711**	+.080 [+.030,+.129]
WebQ	250	.720	.693	**.736**	+.043 [−.009,+.098]
Llama-QA	250	.745	.639	**.757**	+.118 [+.050,+.192]
SD-QA	200	.632	.706	**.736**	+.029 [−.027,+.086]
Reasoning-zh	202	.578	.491	**.606**	+.115 [+.051,+.180]

Table 17: Native-regime gated accuracy (8ad remix: native scores select, native local outcomes + cached always-arm expert outcomes mix). p = permutation vs. matched-rate random; escalation rate in parentheses. Reasoning-zh: the gate never fires (all-English calibration places zh scores below every global threshold).

pool	local	balanced	aggressive	always
frozen test	.371	.492 (24%, p=.001)	.596 (44%, p<.0001)	.667
TriviaQA	.612	.756 (26%, p<.0001)	.872 (64%, p<.0001)	.916
WebQ	.392	.556 (31%, p=.024)	.740 (73%, p=.008)	.816
Llama-QA	.760	.780 (4%, p=.031)	.828 (24%, p=.0035)	.912
SD-QA	.440	.660 (30%, p<.0001)	.830 (70%, p<.0001)	.895
Reasoning-zh	.564	— (0%)	.579 (3%, n.s.)	.807
AlpacaEval (VB 1–5)	3.81	3.83 (4%, n.s.)	4.12 (24%, p=.019)	4.76

was the harness read point, not liveness itself. Thresholds again do *not* transfer (balanced quantile 0.386→0.645); scores are decoding-temperature invariant (0.7-trained probe on temp-0.1 features: AUC 0.843).

Validity. Re-mixing native local outcomes (gpt-judged, deployed decoding) with cached always-arm expert outcomes under the native scores (Table 17): gated accuracy clears matched-rate random on five of six pools — frozen .371→.596 at 44% escalation (p<.0001; ceiling .667), TriviaQA .612→.872 at 64%, WebQ .392→.740, Llama-QA .760→.828, SD-QA .440→.830 at 70% (p<.0001). External fire rates drift from their nominal calib quantiles (aggressive fires 64–73% vs. the nominal 50%) — thresholds are the fragile part of calibration in a third consecutive regime. On Reasoning-zh the gate simply never fires: the all-English calibration places zh scores below every global threshold — the language axis of Appendix H, now expressed as a fire-rate failure rather than an AUC failure. The talker's own local floor is lower in-regime (frozen .483→.371; ≈one third decoding temperature, ≈8 points genuine regime cost), which *raises* the gate's marginal value in deployment. The in-regime calibration scaling curve mirrors the concurrent one and is not saturated at 2,310 rows (internal .717→.830, external mean .643→.709 from 360→2,310).

Live validation. Full live runs on frozen test-240 (real raw-audio ASR uplink, real expert, wait paced at one chunk per second) validate the remix arithmetic where the channel is thin and expose where it is not: balanced delivers .504 at 23% fire (remix predicted .492 at 24%); aggressive delivers .537 at 45% against a .596 remix prediction, and the gap decomposes entirely onto fired turns — live fired-turn accuracy .495–.589 vs. the .667 cached-expert ceiling. The native *relay channel* (the answer re-entering as a text unit and being voiced) is the regime's lossy element — 89% of relay units need the nudge retry — playing the role the transcription uplink played in the turn-based loop; the uplink itself is now clean (raw-audio ASR, p50 0.9 s). Latency, expert-cache warm: expert p50 4.4–5.7 s, wait p50 5–6 chunks, fire→relay first audio p50 ∼7–8 s. Selection is visible live: unfired subsets score .478/.573 against the .371 unconditional local floor.

Floor control, natively. With the harness gone, 8ba's question re-poses on the head itself (108 scripted overlap trials, stims spoken over the talker at fixed offsets). During answer speech, backchannels almost never stop the floor (false-stop 3/24 within a 6-chunk window) with *no* ASR or lexicon anywhere in the loop; sustained question speech yields the floor (6/12 within 6 chunks, median 8.5 post-stim); short burst commands ("Stop!") mostly do not (3/12) — the native head reads sustained speech, not commands, a naturalness-for-reliability trade the old energy-VAD did not make. The escalation machinery is floor-transparent where it should be: relays complete under overlapping stims (26/30), and backchannels during the thinker wait leave the pending relay intact (8/10). The one true interaction: a *new question* during the wait derails the pending relay 7/10 times — the head simply moves on — which is the empirical case for aborting the in-flight expert call on user speech, left as deployment future work. Latency: listen chunks cost 0.04 s and speak chunks $\sim$0.5 s on one H100 (realtime holds); fire$\rightarrow$stall-audio 0.2–1 s; fire$\rightarrow$relay-first-audio is the expert wall clock (13–20 s observed) plus $\sim$2.5 s.